

The Regulatory Intermediation Gap: Evidence from Singapore

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Abstract

When government equity-market interventions operate through private intermediaries, intermediary constraints may favour firms that already meet selection criteria over the intended beneficiaries. We term this divergence the ‘regulatory intermediation gap’. A cross-sectional event study of Singapore’s 2025 equity-market reform (a S\$5 billion capital commitment channelled through professional fund managers and targeting small- and mid-cap firms on the Singapore Exchange (SGX), bundled with deregulation of SGX’s post-listing oversight regime) documents the gap along two dimensions. On the capital-deployment side, the smallest quartile of firms lost 2.5 percent at the bundled announcement while the largest gained 0.2 percent, consistent with fund managers’ investability constraints. The relationship between firm size and returns reversed when the Monetary Authority of Singapore (MAS) later disclosed mandate terms requiring small- and mid-cap allocation. On the deregulation side, high-friction firms were penalised as the market read oversight removal as information loss, while watch-listed firms received a stigma-relief premium, reinforced when SGX subsequently abolished the watch-list. The findings suggest that mandate specificity and targeted burden removal govern whether intermediated equity programmes reach their intended beneficiaries.

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1 Introduction

In February 2025, the Monetary Authority of Singapore (MAS) announced one of the most ambitious equity-market reform packages in its history. That institutional setting matters because MAS is an integrated regulator, charged with developing Singapore as a financial centre while also exercising prudential and market-conduct functions. Front-line supervision of the listed market is delegated to Singapore Exchange Regulation (SGX RegCo), the exchange’s regulatory subsidiary. Drawing on both dimensions of that architecture, the reform advanced on two fronts. On the developmental side, MAS committed S\$5 billion through the Equity Market Development Programme (EQDP) to professional fund managers investing in Singapore Exchange (SGX) equities. That programme was intended to draw institutional capital towards small- and mid-cap firms that had long suffered thin trading and depressed valuations. On the supervisory side, SGX RegCo proposed to loosen the post-listing rules by removing a public financial watch-list, consolidating listing-review functions, and streamlining prospectus and admission requirements. The package therefore joined capital deployment with oversight deregulation in a single announcement aimed at reviving small- and mid-cap listings, with the authorities disclosing the instruments in sequence over the months that followed.

The reform is an instance of a more general regulatory-design problem. Although the decline of small-cap public equities has prompted governments to intervene directly in listed markets, such interventions may first need to pass through an institutional channel before reaching the firms they purport to help. To that end, governments have tended to rely on two instruments: routing public or central-bank capital into listed equities through private intermediaries,¹ or loosening the listing and oversight rules that govern firms once they are public.² The first instrument follows the investability constraints of professional managers; because those managers need securities liquid enough to trade in size, they tend to favour large, liquid stocks. The second instrument withdraws the information that public oversight tools — queries, halts, and suspensions — supplied to investors who cannot easily verify firm quality. The problem is especially acute outside the U.S., where benchmarked funds hold disproportionately more large-cap stock and private securities litigation is a weaker substitute for public oversight. We define the divergence between a policy’s stated target and the firms reached through these channels as the ‘regulatory intermediation gap’. We call the two channels the ‘capital channel’ and the ‘information channel’: the first captures how capital deployment could miss small firms, while the second captures how oversight deregulation could harm previously monitored firms. We proxy the first by firm size and the second by a firm’s accumulated record of regulatory friction.

¹See, e.g., Korea’s Corporate Value-up Programme and the Bank of Japan’s purchases of exchange-traded funds, the latter channelling central-bank capital into index-linked funds: Financial Services Commission (Korea), ‘Active Support to be Provided to Promote Voluntary Efforts of Listed Companies in Enhancing Their Value’, press release, 26 February 2024; Bank of Japan, ‘Comprehensive Monetary Easing’, statement, 5 October 2010.

²See, e.g., the United Kingdom’s overhaul of its Listing Rules and the European Union’s Listing Act, which ease the regulatory burden of being listed: Financial Conduct Authority, *Primary Markets Effectiveness Review: Feedback to CP23/31 and Final UK Listing Rules* (Policy Statement PS24/6, July 2024); Regulation (EU) 2024/2809 of the European Parliament and of the Council of 23 October 2024 amending Regulations (EU) 2017/1129, (EU) No 596/2014 and (EU) No 600/2014 (Listing Act) [2024] OJ L 2024/2809.

Singapore's 2025 package allows us to observe that gap because the February announcement bundled the two instruments, while later announcements activated narrower, partially overlapping components over the following nine months. We ask how the market priced the capital-deployment and oversight-information channels, and which design features changed that pricing, by conducting an event study that compares announcement returns across a common set of firms. Accordingly, what we observe at each event is the market's valuation of the capital and oversight instruments. Whether the combination of developmental and supervisory functions within one regulatory architecture shaped the package's design is a question about regulatory deliberation, which our empirical design cannot answer. Our sequence of event tests, placebo comparisons, and robustness checks nevertheless supports the channel-pricing interpretation over a generic size rotation or deregulation premium.

The first set of findings concerns the capital channel. At the bundled February announcement, the market priced the capital programme as a large-cap event: the smallest quartile of firms recorded cumulative abnormal returns (their stock-price reactions beyond what the broad market would predict) of -2.5 percent, while the largest gained 0.2 percent, a spread of 2.8 percentage points that ran directly against the programme's stated purpose. This is the pattern one would expect when S\$5 billion ($\approx$ US\$3.8 billion) is delegated to managers whose investability constraints favour large, liquid securities. That interpretation is reinforced by the July disclosure, when MAS named the first three EQDP managers, placed an initial S\$1.1 billion ($\approx$ US\$0.8 billion) with them, and disclosed mandate terms that specified significant allocation to small- and mid-cap equities. The size gradient (the cross-sectional relationship between firm size and returns) reversed, with smaller firms gaining more than larger ones, and the average firm posted a positive abnormal return of roughly three percent. Cumulated across the two events, the gap narrowed without fully closing. Mandate specificity — that is, how tightly the mandate dictated where managers had to invest — therefore offset the intermediation bias without eliminating it.

The second set of findings concerns the information channel. We find that the market priced deregulation as a loss of information for firms most exposed to oversight. A firm's accumulated regulatory friction is its record of regulatory attention over time, comprising the prior SGX queries (public questions the exchange puts to an issuer, typically about unusual trading), trading halts, and suspensions it had drawn, together with rarer formal regulatory actions. Firms carrying more friction recorded lower announcement returns, with each additional friction event predicting roughly a third of a percentage point of decline even after controlling for size. Splitting the friction measure by type and by age locates the penalty in halts and suspensions and in older friction events; routine queries and recent scrutiny show no detectable effect. The market thus treated regulatory attention as a stock of reputation rather than a flow of current monitoring. Once the public query regime is withdrawn, investors could no longer tell which firms the regulator has flagged, and an adverse-selection problem reappears across the previously monitored segment.³

The watch-list is an exception to this pattern and reveals that the information effects of deregulation depend on the design of the regulatory instrument withdrawn. Holding a firm's overall friction constant, firms on SGX's financial watch-list posted positive announcement returns; when

³George A Akerlof, 'The Market for "Lemons": Quality Uncertainty and the Market Mechanism' (1970) 84 Q J Econ 488

SGX RegCo abolished the watch-list in October, those previously watch-listed firms gained sharply relative to others. The watch-list label imposed financing and counterparty costs that outweighed whatever information it conveyed, so the market priced its removal as relief from stigma even as it priced the removal of genuinely informative monitoring as a loss. The two responses—the friction penalty and the watch-list relief—are consistent because the market’s reaction depended more on the type of regulatory instrument withdrawn than on the overall amount of oversight removed. The friction results provide the primary evidence for this information channel, while the smaller watch-listed sample offers corroborating evidence.

Our findings point to a design lesson for regulators considering comparable programmes: whether an intervention reaches its intended firms may depend more on the terms of delegation and the selectivity of burden removal than on the scale of the commitment. We develop this lesson through three contributions. First, we document the regulatory intermediation gap in public equity markets, extending to listed equities a concern that earlier work studied mainly in directed lending and government procurement, where the agents who deploy public capital shape who receives it. Second, we show that the gap responds to regulatory design: the reversal across firms between the February announcement and the July mandate disclosure indicates that specifying mandate terms changes how intermediated capital is priced for the target segment. Third, we bring the informational theory of securities regulation to bear on deregulation by showing that the market prices the withdrawal of informative monitoring and the removal of a stigmatising label in opposite directions. The bundled February announcement and its staggered unbundling also supply the cross-event variation that our analysis exploits, although the separation across later events is only partial. Our empirical design therefore rests mainly on comparing firms within the February announcement and on the February-to-July reversal; [Section 4](#) explains the resulting limits on attribution.

The remainder of our article proceeds as follows. [Section 2](#) develops the intermediation gap as a problem of regulatory design and places the two channels in the relevant literatures. [Section 3](#) describes Singapore’s equity market, the integrated regulatory architecture, and the reform timeline. In [Section 4](#), the data and the empirical approach are set out, including the limits of what our design can establish. [Section 5](#) presents the findings on the two channels and how they behaved across events. [Section 6](#) draws out the implications for regulatory design, while [Section 7](#) concludes.

2 The Intermediation Gap as a Regulatory Design Problem

Although governments seeking to support underserved equity segments can usually articulate their policy objectives with reasonable precision, they seldom purchase shares directly in the market. Instead, they typically work through two delegated instruments: channelling public capital through private fund managers and adjusting the rules by which firms are monitored once they list. The resulting ‘regulatory intermediation gap’ is a problem of institutional design. Each delegation places the policy objective in the hands of an intermediary whose own constraints may pull the intervention away from its stated purpose. On the capital channel, public money

passes through professional managers whose mandates, fee structures, and evaluation against a benchmark index tend to favour large and liquid securities; consequently, it may drift towards firms that already satisfy ordinary investability criteria. On the information channel, investors use visible public instruments — queries, halts, suspensions, and classifications — to price firms whose quality is difficult to verify. Withdrawing those instruments therefore removes information as well as regulatory burden. In each channel, the size of the gap depends on how specifically the intervention is designed. This section develops that point through two design choices: how specifically the mandate governing delegated capital is framed, and which oversight instrument is withdrawn.

2.1 The Capital Channel

Public capital often reaches its statutory or policy target through private agents. The central question, therefore, is whether those agents can reliably transmit the policy intent despite interference from their own incentive calculus. This slippage creates an ‘intermediation gap’, whereby capital committed to a public objective may be redirected by private intermediaries towards other ends. Granja et al. provide a clear illustration from the U.S. Paycheck Protection Programme, showing that bank-level differences in disbursement caused funds to flow initially towards regions less affected by the pandemic.⁴ The point extends beyond crisis lending. Even without bad faith, the U.S. Community Reinvestment Act 1977 shows how a directed-credit obligation could miss the communities it names. The Act encouraged banks to lend in low-income communities by tying favourable assessments to approval of mergers and expansion. In practice, banks tended to optimise towards the assessment criteria themselves, concentrating lending around examination dates⁵ and received inflated grades that left roughly 96 percent of banks rated satisfactory or better.⁶ Little evidence emerged, however, that credit had shifted towards lower-income communities.⁷ These examples suggest that intermediation is more likely to preserve policy intent when intermediaries’ incentives and the terms governing their mandates are aligned with the public objective.

Equity-market delegation operates through a chain of intermediaries whose influence over investment decisions has long attracted policy scrutiny. The Kay review of UK equity markets diagnosed a misalignment along the investment chain, in which participants came to evaluate market performance through the perspective of intermediaries rather than through the needs of companies or end investors.⁸ That diagnosis built on earlier official assessments: the Myners

⁴João Granja and others, ‘Did the paycheck protection program hit the target?’ (2022) 145 J Fin Econ 725

⁵Sumit Agarwal, Efraim Benmelech, Nittai Bergman and Amit Seru, ‘Did the Community Reinvestment Act (CRA) Lead to Risky Lending?’ (NBER Working Paper No 18609, 2012).

⁶Peter Conti-Brown and Brian D Feinstein, ‘Banking on a Curve: How to Restore the Community Reinvestment Act’ (2023) 13 Harv Bus L Rev 335.

⁷Robert B Avery, Paul S Calem and Glenn B Canner, ‘The Effects of the Community Reinvestment Act on Local Communities’ (Federal Reserve Board, Division of Research and Statistics, March 2003); Michelle Minton, ‘The Community Reinvestment Act’s Harmful Legacy: How It Hampers Access to Credit’ (Competitive Enterprise Institute, No 132, March 2008).

⁸John Kay, ‘The Kay Review of UK Equity Markets and Long-Term Decision Making’ (Department for Business,

review documented misaligned incentives in institutional investment decision making,⁹ while the Walker review identified governance failures in the financial institutions that stand between savers and firms.¹⁰ Braun and Christophers describe the same mechanism at market scale, observing that where a small number of large managers hold most shares, intermediary constraints tend to dominate end-investor interests.¹¹ Among the largest index managers, Bebchuk and Hirst document the substantial and growing voting power of the dominant fund families, together with the agency distortions produced by delegation,¹² including under-investment in stewardship and deference to portfolio-company management.¹³ These concerns extend beyond stewardship because intermediary incentives also shape the allocation of capital across firms.

These allocation effects are especially important given the structural decline in small-cap public equity. Across developed markets, the number of listed firms has fallen; in the U.S., it has roughly halved from its late-1990s peak,¹⁴ driven substantially by the collapse in IPO activity among small firms.¹⁵ Concentration has also risen among the firms that remain, as industries have consolidated following the disappearance of smaller public competitors and as a growing share of institutional capital has moved into the largest firms.¹⁶

Benchmarking may reinforce this shift. Because investors cannot directly observe a manager's skill, compensation is often tied to performance against a benchmark, which encourages managers to remain close to the index.¹⁷ Institutions accordingly tend to overweight securities in their benchmarks,¹⁸ particularly when managers track those benchmarks closely.¹⁹ When many managers face these incentives, demand shifts away from securities outside the index and towards benchmark

Innovation and Skills 2012)

⁹Paul Myners, 'Institutional Investment in the United Kingdom: A Review' (HM Treasury 2001)

¹⁰David Walker, 'A Review of Corporate Governance in UK Banks and Other Financial Industry Entities' (HM Treasury 2009)

¹¹Benjamin Braun and Brett Christophers, 'Asset Manager Capitalism: An Introduction to Its Political Economy and Economic Geography' (2024) 56 *Env & Plan A* 546

¹²Lucian Bebchuk and Scott Hirst, 'Index Funds and the Future of Corporate Governance: Theory, Evidence, and Policy' (2019) 119 *Colum L Rev* 2029

¹³See also Bebchuk and Hirst, 'Big three power, and why it matters' (2022) 102 *BU L Rev* 1547

¹⁴Craig Doidge, G Andrew Karolyi and René M Stulz, 'The U.S. Listing Gap' (2017) 123 *J Fin Econ* 464; Kathleen M Kahle and René M Stulz, 'Is the US Public Corporation in Trouble?' (2017) 31 *J Econ Persp* 67; Mark J Roe and Charles CY Wang, 'Half the Firms, Double the Profits: Public Firms' Transformation, 1996–2022' (2025) 8 *J L Fin & Acct* 211; Craig Doidge and others, 'Are there too few publicly listed firms in the US?' (2025) 60 *Fin Rev* 317

¹⁵Xiaohui Gao, Jay R Ritter and Zhongyan Zhu, 'Where Have All the IPOs Gone?' (2013) 48 *J Fin & Quant Anal* 1663

¹⁶Gustavo Grullon, Yelena Larkin and Roni Michaely, 'Are US Industries Becoming More Concentrated?' (2019) 23 *Rev Fin* 697; Spencer Y Kwon, Yueran Ma and Kaspar Zimmermann, '100 years of rising corporate concentration' (2024) 114 *Am Econ Rev* 2111

¹⁷Zhiguo He and Wei Xiong, 'Delegated Asset Management, Investment Mandates, and Capital Immobility' (2013) 107 *J Fin Econ* 239

¹⁸James A Bennett, Richard W Sias and Laura T Starks, 'Greener Pastures and the Impact of Dynamic Institutional Preferences' (2003) 16 *Rev Fin Stud* 1203; Anna Pavlova and Taisiya Sikorskaya, 'Benchmarking intensity' (2023) 36 *Rev Fin Stud* 859

¹⁹KJ Martijn Cremers and Antti Petajisto, 'How Active Is Your Fund Manager? A New Measure That Predicts Performance' (2009) 22 *Rev Fin Stud* 3329; Martijn Cremers and others, 'Indexing and active fund management: International evidence' (2016) 120 *J Fin Econ* 539

constituents, raising their relative prices.²⁰ As more capital is managed against benchmarks, firms' cost of capital may reflect their index membership and weight beyond their underlying fundamentals.²¹ Index-linked demand may also weaken the incorporation of firm-specific information into prices²² and disproportionately inflate mega-cap valuations.²³ Similar allocation effects appear even among long-horizon institutions whose mandates might otherwise be expected to support firm-specific valuation.²⁴ A public programme targeting the small-cap segment therefore operates against the prevailing direction of intermediated flows. The reform we study makes this tension observable in market prices, as the reaction to a S\$5 billion programme channelled through fund managers reflects expectations about how intermediary incentives will shape the allocation of capital.

As exchanges compete more intensely to attract and retain firms amid a decline in public listings, governments in several jurisdictions have adopted programmes addressing similar concerns.²⁵ These programmes differ along two dimensions that matter for our article: the channel through which public policy reaches the market and the degree to which the intervention targets an underserved segment. Public-private capital initiatives such as China's Government Guidance Funds route state money through appointed managers, although they operate mainly through venture-capital and private-equity channels rather than through secondary-market purchases of listed equities.²⁶ They therefore reveal relatively little about how managers in public equity markets respond to the terms of a mandate. Supply-side reforms instead lower the barriers to listing. The European Union's Listing Act, in force from December 2024, and the United Kingdom's 2024 overhaul of its Listing Rules both ease the requirements for admission and continued listing.²⁷ These reforms operate on the stock of listings rather than on the allocation of capital among firms already listed, and their lasting effects remain difficult to assess.²⁸ Korea's Corporate Value-up Programme combines issuer disclosure, governance incentives, and index-linked infrastructure. Although the programme was associated with favourable headline performance, its wider market effects have been uneven and concentrated among larger firms.²⁹

²⁰Andrea Buffa, Dimitri Vayanos and Paul Woolley, 'Asset Management Contracts and Equilibrium Prices' (Working Paper, 2014)

²¹Anil K Kashyap and others, 'The benchmark inclusion subsidy' (2021) 142 J Fin Econ 756

²²Jeffrey Wurgler, 'On the Economic Consequences of Index-Linked Investing' in William T Allen and others (eds), *Challenges to Business in the Twenty-First Century: The Way Forward* (American Academy of Arts and Sciences 2010); Doron Israeli, Charles MC Lee and Suhas A Sridharan, 'Is There a Dark Side to Exchange Traded Funds? An Information Perspective' (2017) 22 Rev Acct Stud 1048

²³H Jiang, D Vayanos and L Zheng, 'Passive investing and the rise of mega-firms' (2025) 38 Rev Fin Stud 3461

²⁴Aleksandar Andonov, 'Delegated Investment Management in Alternative Assets' (2022) 13 Rev Corp Fin Stud 264

²⁵Curtis J Milhaupt and Wolf-Georg Ringe, 'The Political Economy of Global Stock Exchange Competition' (ECGI Law Working Paper No 872/2025, 2025)

²⁶Xinfei Huang, Yue Zhang and Zhe Zong, 'Governmental Venture Capital and Investor Sentiment: Evidence from Chinese Government Guidance Funds' (2025) 99 J Intl Fin Mkts Inst & Money 102120; for the funds' governing framework, see Ministry of Finance (China), Interim Measures for the Administration of Government Investment Funds (Cai Yu [2015] No 210, 12 November 2015)

²⁷Listing Act (n 2); Financial Conduct Authority, PS24/6 (n 2)

²⁸Brian R Cheffins and Bobby V Reddy, 'Law and Stock Market Development in the UK over Time: An Uneasy Match' (2023) 43 OJLS 725

²⁹Bora Lee and Kian Heng Peh, 'Narrowing the Korea Discount: Stock Market Reform via Corporate Value-up' (AMRO Blog, 30 March 2026); for the programme's terms, see Financial Services Commission (Korea) (n 1)

The closest precedent for channelling public capital into listed equities is the Bank of Japan's exchange-traded-fund purchase programme, which deployed central-bank funds through a trust-bank structure into index-linked funds.³⁰ Because that intervention pursued monetary-policy objectives through index replication rather than active mandates directed at an underserved segment, its benefits accrued to issuers already included in the relevant indices.

These programmes engage the capital channel, but do not provide variation in mandate specificity that would isolate its role in the intermediation gap. The Singapore reform supplies that variation: it first commits capital for deployment through active managers and later discloses mandate terms specifying significant allocation to small- and mid-cap equities. Existing studies document where delegated capital goes but, to the best of our knowledge, provide little direct empirical evidence on whether mandate terms alter its allocation in public equity markets. The announcement returns around the reform bear directly on that question. They are consistent with an intermediation gap in a listed-equity setting, while the reversal between the February reform announcement (E1) and the July mandate disclosure (E3) indicates that greater mandate specificity attenuates that gap.³¹

2.2 The Information Channel

A longstanding debate asks whether securities regulation creates or destroys firm value. It rests on a broader literature showing that legal institutions and investor protection shape financial development³² and that the design of securities laws, especially their disclosure and liability provisions, governs market outcomes.³³ The two sides of that debate imply opposite predictions for how the market should price the removal of oversight. One account emphasises the costs imposed on regulated firms and therefore predicts that deregulation will increase their value. Public-choice theories associated with Stigler³⁴ and Peltzman³⁵ supply its theoretical foundation. Event studies of the Sarbanes–Oxley Act likewise document substantial compliance costs concentrated in smaller firms,³⁶ and issuers exempted from disclosure obligations under the JOBS Act experienced positive returns.³⁷ Even within this account, a credible commitment to stricter oversight may reduce

³⁰Ben Charoenwong, Randall Morck and Yupana Wiwattanakantang, 'Bank of Japan Equity Purchases: The Final Frontier in Extreme Quantitative Easing' (NBER Working Paper No 25525, 2019); Mitsuru Katagiri, Junnosuke Shino and Koji Takahashi, 'Bank of Japan's ETF Purchase Program and Equity Risk Premium: A CAPM Interpretation' (2025) 73 J Fin Mkts 100961; for the programme's terms, see Bank of Japan (n 1)

³¹The full reform timeline appears in [Table 1](#); the inferential backbone of the reversal, including the stacked interaction estimate and the paired-reversal placebo, accompanies the size exhibit in [Section 5](#).

³²Rafael La Porta and others, 'Legal Determinants of External Finance' (1997) 52 J Fin 1131; Rafael La Porta and others, 'Law and Finance' (1998) 106 J Pol Econ 1113; Andrei Shleifer and Daniel Wolfenzon, 'Investor Protection and Equity Markets' (2002) 66 J Fin Econ 3

³³Rafael La Porta, Florencio Lopez-de-Silanes and Andrei Shleifer, 'What Works in Securities Laws?' (2006) 61 J Fin 1

³⁴George J Stigler, 'The Theory of Economic Regulation' (1971) 2 Bell J Econ & Mgmt Sci 3

³⁵Sam Peltzman, 'Toward a More General Theory of Regulation' (1976) 19 J L & Econ 211

³⁶Ivy Xiyang Zhang, 'Economic Consequences of the Sarbanes–Oxley Act of 2002' (2007) 44 J Acct & Econ 74; Peter Iliev, 'The Effect of SOX Section 404: Costs, Earnings Quality, and Stock Prices' (2010) 65 J Fin 1163

³⁷Dharmika Dharmapala and Vikramaditya S Khanna, 'The Costs and Benefits of Mandatory Securities Regulation: Evidence from Market Reactions to the JOBS Act of 2012' (SSRN Working Paper, 2014)

adverse selection and thereby lower firms' cost of capital.³⁸

A second account treats securities regulation as informational infrastructure that lowers the cost of producing accurate prices. Mandatory disclosure, fraud rules, and the monitoring instruments that accompany them support the production and incorporation of information into prices through the arbitrage mechanisms catalogued by Gilson and Kraakman.³⁹ Withdrawing secondary-market oversight should therefore be priced negatively for firms whose quality investors find difficult to verify independently.⁴⁰ The disclosure-economics literature supplies much of the evidence for this view. Fuller disclosure reduces information asymmetry and the cost of capital,⁴¹ and disclosure mandates have generated positive abnormal returns and improvements in liquidity in several settings.⁴² The intensity and credibility of enforcement also matter because investors price the expected effectiveness of oversight rather than its statutory form alone.⁴³ In the opposite direction, event studies find negative abnormal returns for firms affected by the withdrawal of mandated reporting or disclosure oversight.⁴⁴

Disclosure regimes translate these competing accounts into institutional form. In Singapore, officials historically invoked 'caveat emptor', assigning investors responsibility for assessing issuer quality once the prescribed information had been disclosed.⁴⁵ The original justification for the U.S. federal securities laws placed greater emphasis on verification by issuers and the institutions bringing securities to market, with President Roosevelt describing disclosure as a means of ensuring that 'the seller also beware'.⁴⁶ While both are disclosure regimes, they differ in how they allocate verification responsibility. Where public oversight supplements issuer disclosure, the queries, halts, and suspensions it produces give investors information about firms that have

³⁸Cécile Carpentier, Douglas Cumming and Jean-Marc Suret, 'The Value of Capital Market Regulation: IPOs versus Reverse Mergers' (2012) 9 J Empirical Legal Stud 56

³⁹Ronald J Gilson and Reinier H Kraakman, 'The Mechanisms of Market Efficiency' (1984) 70 Va L Rev 549

⁴⁰Zohar Goshen and Gideon Parchomovsky, 'The Essential Role of Securities Regulation' (2006) 55 Duke LJ 711

⁴¹Paul M Healy and Krishna G Palepu, 'Information Asymmetry, Corporate Disclosure, and the Capital Markets: A Review of the Empirical Disclosure Literature' (2001) 31 J Acct & Econ 405; Christian Leuz and Robert E Verrecchia, 'The Economic Consequences of Increased Disclosure' (2000) 38 J Acct Res 91; Christian Leuz and Peter D Wysocki, 'The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research' (2016) 54 J Acct Res 525; Hans B Christensen, Luzi Hail and Christian Leuz, 'Capital-Market Effects of Securities Regulation: Prior Conditions, Implementation, and Enforcement' (2016) 29 Rev Fin Stud 2885

⁴²Brian J Bushee and Christian Leuz, 'Economic Consequences of SEC Disclosure Regulation: Evidence from the OTC Bulletin Board' (2005) 39 J Acct & Econ 233; Michael Greenstone, Paul Oyer and Annette Vissing-Jorgensen, 'Mandated Disclosure, Stock Returns, and the 1964 Securities Acts Amendments' (Working Paper, 2005); Allen Ferrell, 'Mandatory Disclosure and Stock Returns: Evidence from the Over-the-Counter Market' (2007) 36 J Legal Stud 213

⁴³John C Coffee, 'Law and the Market: The Impact of Enforcement' (SSRN Working Paper, 2007); Hans B Christensen, Luzi Hail and Christian Leuz, 'Mandatory IFRS Reporting and Changes in Enforcement' (2013) 56 J Acct & Econ 147

⁴⁴See Nuno Fernandes, Ugur Lel and Darius P Miller, 'Escape from New York: The Market Impact of Loosening Disclosure Requirements' (2010) 95 J Fin Econ 129; Vanessa Behrmann and others, 'The Deregulation of Quarterly Reporting and Its Effects on Information Asymmetry and Firm Value' (2025) 64 Rev Quant Fin & Acct 1221; Jörg-Markus Hitz and Florian Moritz, 'Turning Back the Clock on Disclosure Regulation? Evidence from the Termination of the Quarterly Reporting Mandate in Europe' (SSRN Working Paper, 2019)

⁴⁵Lee Hsien Loong, Deputy Prime Minister, *Investor Education & New Framework for Financial Advisers*, speech at the 1st Anniversary Dinner of the Securities Investors Association (Singapore), 7 July 2000, <https://www.mas.gov.sg/news/speeches/2000/investor-education-new-framework-for-financial-advisers--07-jul-2000>.

⁴⁶Franklin D. Roosevelt, Message to Congress on Federal Supervision of Investment Securities, 29 March 1933.

attracted scrutiny. Reducing that oversight transfers more of the verification burden to investors.

A market that prices this transfer should penalise the firms for which public monitoring had supplied information that investors cannot readily reproduce. Once the public regime ceases to generate that flow of signals, the condition of previously scrutinised firms becomes difficult to observe, and reduced observability could revive the adverse-selection problem Akerlof described by making sound and unsound issuers harder to distinguish.⁴⁷ Public enforcement may also generate positive valuation spillovers by reassuring investors about the credibility of oversight,⁴⁸ so its withdrawal removes reassurance as well as firm-specific information. Compliance costs and informational benefits therefore jointly determine how the market prices the removal of a given instrument.⁴⁹

Work on regulatory stigma explains how these accounts can coexist, as the regulatory instrument itself may impose costs beyond the information it conveys. Regulatory sanctions may inflict reputational damage greater than the direct financial penalty—damage that a visible classification could transmit regardless of the underlying conduct that triggered it.⁵⁰ Emergency central-bank lending supplies a close analogy. During the 2007–2008 crisis, banks avoided borrowing from the Federal Reserve’s discount window (its standing source of short-term loans) because use of the facility itself could signal financial distress, so the facility’s stigma harmed the institutions it was designed to assist.⁵¹ SGX’s financial watch-list worked through a comparable mechanism, with the label impairing financing and making counterparties reluctant to deal. Removal of the label could therefore be priced as stigma relief, even as removal of genuinely informative monitoring was priced as a loss. Seen this way, the watch-list separates two functions that broad deregulation tends to bundle together: the punitive function of a stigmatising label and the informational function of active monitoring. The institutional design problem is to make burden removal sufficiently selective that it retires the former while preserving the latter.⁵² A parallel form of sorting appears across listing regimes. Nielsson finds that London’s AIM,⁵³ with its lighter listing requirements, allowed smaller firms to raise capital without attracting observably lower-quality issuers; issuers sorted between exchanges by size and sensitivity to regulatory cost. Singapore’s deregulation penalty followed the same dimensions, falling on its smaller and higher-friction firms. Our friction results provide the primary evidence on this channel, with the watch-list findings supplying corroborating evidence.

⁴⁷Akerlof (n 3); John C Coffee Jr, ‘Market Failure and the Economic Case for a Mandatory Disclosure System’ (1984) 70 Va L Rev 717

⁴⁸Roger Silvers, ‘The Valuation Impact of SEC Enforcement Actions on Nontarget Foreign Firms’ (2016) 54 J Acct Res 187

⁴⁹J Lawrence, ‘The Economics of Market Confidence: (Ac)Costing Securities Market Regulations’ (2000) 17 Co & Sec LJ 171

⁵⁰John Armour, Colin Mayer and Andrea Polo, ‘Regulatory Sanctions and Reputational Damage in Financial Markets’ (SSRN Working Paper, 2010); John Armour, Colin Mayer and Andrea Polo, ‘Naming and Shaming: Evidence from Event Studies’ in *The Cambridge Handbook of Compliance* (Cambridge University Press 2019)

⁵¹Olivier Armandier and others, ‘Discount Window Stigma During the 2007–2008 Financial Crisis’ (2015) 118 J Fin Econ 317; Sriya Anbil, ‘Managing Stigma During a Financial Crisis’ (2018) 130 J Fin Econ 166

⁵²Omri Ben-Shahar and Carl E Schneider, *More Than You Wanted to Know: The Failure of Mandated Disclosure* (Princeton University Press 2014) 194–95.

⁵³Ulf Nielsson, ‘Do less regulated markets attract lower quality firms? Evidence from the London AIM market’ (SSRN Working Paper, 2013)

3 Institutional Background

Singapore's 2025 equity-market reform emerged from a regulatory architecture in which one statutory authority carries a developmental mandate alongside its supervisory one. MAS, the country's integrated regulator for banking, insurance, and securities, is also charged by its enabling legislation with developing Singapore as a financial centre.⁵⁴ Front-line oversight of the listed market is delegated to SGX RegCo, the exchange's independent regulatory subsidiary, which administers the listing rules, market surveillance, and enforcement under MAS's statutory supervision. This allocation matters for the reform's institutional sequence, which [Table 1](#) maps event by event: because the developmental and supervisory instruments were issued by different bodies under different legal authorities, capital deployment and oversight recalibration could be announced together and then separated over the following months.

⁵⁴Monetary Authority of Singapore Act 1970 (2020 Rev Ed) s 4(2)(d) (mandate to develop Singapore as a financial centre) and s 4(2)(b) (integrated supervision of the financial sector and financial-stability surveillance); see generally Hans Tjio, Wai Yee Wan and Kwok Hon Yee, *Principles and Practice of Securities Regulation in Singapore* (3rd edn, LexisNexis 2017).

Table 1: Timeline, Acting Authority, and Pricing Channels of Singapore’s 2025 Equity-Market Reform

Event	Date	Acting body	Instrument	Statutory / regulatory basis	Primary channel
E0	2 Aug 2024	MAS	Formation of the Equities Market Review Group (no substantive measures)	MAS developmental mandate (MAS Act s 4(2)(d))	Signal (no measures)
E1	21 Feb 2025	MAS	S\$5bn EQDP capital commitment; investor-participation measures (GIP Option C, GEMS, tax exemptions)	MAS developmental mandate; EQDP capital commitment (MAS Act s 4(2)(d), 13B, 23(1)(m), 128(1)(a), 129)	Capital
(E1)	(cont.)	SGX RegCo	Proposed disclosure-based regime: listing-review consolidation, streamlined admission and prospectus requirements, watch-list removal proposal	Delegated listing-regulation authority; proposed delegation of prospectus lodgment / registration (SFA s 320(1))	Information
E2	15 May 2025	SGX RegCo	Consultation paper on a more disclosure-based regime (watch-list removal, query-to-engagement shift, narrowed suspension criteria)	Delegated listing-regulation authority	Information
E3	21 Jul 2025	MAS	Appointment of first three EQDP managers; S\$1.1bn placed with managers; disclosure of small/mid-cap mandate terms; investor-recourse proposals	MAS developmental mandate; EQDP mandate terms (MAS media release, 21 Jul 2025)	Capital
E4	29 Oct 2025	SGX RegCo	Watch-list abolition; cessation of public trading queries (replaced by private engagement); listing-review rule changes	Delegated listing-regulation authority; listing-rule amendment power	Information (implementation)
E5	19 Nov 2025	MAS	Final Review Group report; second batch of six EQDP managers (S\$2.85bn); SGX–Nasdaq dual-listing bridge	MAS developmental mandate; EQDP capital commitment (MAS Act s 4(2)(d), 13B, 23(1)(m), 128(1)(a), 129)	Capital
(E5)	(cont.)	SGX RegCo	Market-structure measures (board-lot reduction, designated-market-maker incentives)	Exchange market-operation and listing-rule authority	Market structure

Notes: The table lists each reform event with its date, acting body, principal instrument, statutory or regulatory basis, and the pricing channel it primarily activates. It distinguishes the developmental instruments, committed by the Monetary Authority of Singapore (MAS) under its developmental mandate and placed with appointed managers (the EQDP capital at E1 and the manager appointments and mandate disclosure at E3 and E5), from the supervisory instruments, issued by Singapore Exchange Regulation (SGX RegCo) under its delegated listing-regulation authority and MAS’s statutory oversight (the consultation, deregulation, and watch-list abolition at E1, E2, E4, and E5). The “Primary channel” column records whether each event activates the capital channel (capital deployment through professional fund managers) or the information channel (deregulation of SGX’s oversight regime); E1 bundles both, shown as its two sub-rows, while subsequent events progressively unbundle them, and the market-structure measures at E5 (board-lot reduction, designated-market-maker incentives) operate outside the two channels. The Equities Market Review Group (E0, E1, E5) was chaired by the Deputy Chairman of MAS. E0 is excluded from hypothesis testing.

Gurrea-Martínez⁵⁵ provides the closest precedent for our study through a comprehensive qualitative assessment that places the 2025 reform within a comparative account of market-competitiveness initiatives. That account identifies the principal policy instruments and their likely effects, but does not examine how the market priced the reform. Singapore is especially well suited to such an inquiry for two reasons. First, concentrated ownership and limited free float may cause intermediary investability constraints to bind more tightly than in deeper markets. Second, Singapore relies heavily on public oversight and offers relatively few private-enforcement substitutes, so the withdrawal of public regulatory instruments may leave few offsetting mechanisms. The institutional foundations of these features, including the financial watch-list and the post-listing oversight framework, are developed in the subsections that follow.

3.1 The Listing Drain and a History of Regulatory Tightening

Although Singapore has become one of the region's largest centres for private-capital formation through private equity and venture capital, SGX continues to combine a credible regulatory reputation with persistently weak listing activity and trading interest. The depth of Singapore's private-capital ecosystem suggests that the decline in listings cannot be attributed simply to a shortage of viable enterprises. Instead, the 'listing drain' appears to reflect the limited attractiveness of the public market as a financing and trading venue. Over the past decade, market participants and policymakers have repeatedly expressed concern about the decline in initial public offerings, falling trading volumes, and the stagnation of market capitalisation relative to Hong Kong, Tokyo, and Sydney.⁵⁶ These pressures are most acute among small- and mid-cap equities. Thin analyst coverage limits investor awareness, which tends to reduce institutional participation, depress valuations, widen bid-ask spreads, and raise transaction costs. In turn, those transaction costs deter the coverage that might have drawn institutions back. The same circular pressure typically repels new listings.⁵⁷

These wide spreads are a symptom of a thin market, where many counters (listed securities) trade only intermittently and quoted depth remains shallow even for firms of moderate size.⁵⁸ Claessens and others situate the development of a small exchange within the broader internationalisation of stock markets,⁵⁹ in which order flow and listings migrate towards the deepest venues. In light of

⁵⁵Aurelio Gurrea-Martínez, 'Strengthening the International Competitiveness of Capital Markets: Global Insights and Local Strategies' (SMU School of Law Research Paper No 18/2025, 2025)

⁵⁶Owen Walker, 'Singapore's Stock Exchange Hits 20-Year Low in Listed Companies', *Financial Times*, 27 December 2024, <https://www.ft.com/content/bd40562e-1414-485e-9a3a-b2980d7a3d4e>.

⁵⁷The structural sources of this pattern are developed in Section 2.1; our own evidence on the size distribution and the analyst-coverage cliff appears in Figures 1 and 5. The two listing boards differ in admission stringency: the Mainboard imposes quantitative criteria including a minimum pre-tax-profit threshold (S\$30 million, reduced to S\$10 million in October 2025) and market-capitalisation requirements, while Catalist, established in 2007, operates a more flexible sponsor-supervised regime for smaller growth companies that has nonetheless attracted few high-quality listings, with many constituents trading infrequently at deep discounts.

⁵⁸Murphy Jun Jie Lee, 'The Microstructure of Trading Processes on the Singapore Exchange' (PhD thesis, University of Technology Sydney 2013)

⁵⁹Stijn Claessens, Daniela Klingebiel and Sergio L Schmukler, 'Stock Market Development and Internationalization: Do Economic Fundamentals Spur Both Similarly?' (2006) 13 *J Empirical Fin* 316

this, a reform that seeks to draw capital towards the lower tiers of the market works against the general tendency of order flow to concentrate in the largest, most liquid names.

The reform we study also followed a prolonged period of regulatory discipline. Until the 1985 market crisis, when the manipulation of several large listed companies forced a three-day exchange closure, formal securities regulation imposed minimal substantive constraints. The legislative and supervisory response produced a sustained tightening through the 1997 Asian Financial Crisis, even as Western markets were generally moving in a deregulatory direction over the same period.⁶⁰ Because Singapore's corporate and securities law has historically turned such market episodes into prompt rule changes, reform announcements may carry credible information content about near-term legal change.⁶¹

By the time the 2025 measures were announced, SGX had operated under comparatively stringent post-listing oversight for roughly seventeen years, dating from the 2008 introduction of the financial watch-list.⁶² That period of vigilance coincided with a marked fall in listed companies, from a peak of 782 in January 2011 to 622 by July 2024, as weaker firms were delisted, suspended, or forced out through the watch-list process. The 2025 measures are therefore more accurately described as 're-liberalisation', arriving with the regulatory goodwill that the earlier discipline had built up.

3.2 Concentrated Ownership and the Investability Constraint

Because family control and government-linked companies are common among SGX-listed issuers, the effective 'free float' (i.e., shares available to outside institutions) is correspondingly reduced.⁶³ The legal literature on Singapore's listed sector has traced how these controlling stakes bear on governance outcomes and minority-shareholder protection,⁶⁴ the determinants of ownership and board structure,⁶⁵ and board composition and independent directors.⁶⁶ Given that ownership structure, a low free float limits how much a fund manager can acquire without moving the price, thereby reinforcing the tilt of intermediated capital towards larger and more widely held issuers. Passive ownership shows a similar tilt. In the U.S., index funds often hold proportionally more mid-cap than large-cap stock because of their inclusion thresholds, whereas small-caps in Asian and European markets receive little index exposure owing to thin liquidity, transaction costs, and

⁶⁰Tjio and others (n 54)

⁶¹Vincent Ooi and Cheng Han Tan, 'Singapore Company Law and the economy: reciprocal influence over 50 years' (2019) 27 Asia Pac L Rev 14

⁶²Tjio and others (n 54)

⁶³Cheng Han Tan, Dan W Puchniak and Umakanth Varottil, 'State-owned enterprises in Singapore: Historical insights into a potential model for reform' (2014) 28 Colum J Asian L 61

⁶⁴Ernest WK Lim, 'Concentrated ownership, state-owned enterprises and corporate governance' (2021) 41 OJLS 663

⁶⁵YT Mak and Yuan Li, 'Determinants of Corporate Ownership and Board Structure: Evidence from Singapore' (2001) 7 J Corp Fin 235

⁶⁶Ernest Lim, *A case for shareholders' fiduciary duties in common law Asia* (Cambridge University Press 2019); Dan W Puchniak and Luh Luh Lan, 'Independent Directors in Singapore: Puzzling Compliance Requiring Explanation' (2017) 65 Am J Comp L 265

concentrated existing ownership.⁶⁷

3.3 The Watch-List as a Rule-Based Classification

The exchange introduced the ‘financial watch-list’ in 2008 as a rule-based classification governed by specified entry and exit criteria. A Mainboard issuer was placed on the list once it satisfied two objective conditions: it had recorded pre-tax losses for three or more consecutive financial years, and its average daily market capitalisation had fallen below S\$40 million over the preceding six months.⁶⁸ An alternative criterion, based on a volume-weighted average price below twenty cents, operated from March 2015 until June 2020. To be removed from the list, an issuer had to satisfy the same thresholds in the opposite direction. Entry and exit thus turned on objective criteria rather than regulatory judgment, but the list was unusually visible compared with continuing-listing standards elsewhere, and placement on it signalled financial distress in a market where loss-making companies are seldom funded through public equity.⁶⁹

In practice, watch-list status carried substantial stigma. Lenders, counterparties, and investors commonly read the status as a signal of impending delisting risk, impairing the firm’s capacity to raise finance, attract new investment, or restructure. Within two years of the list’s introduction, five of its six inaugural constituents had been delisted.⁷⁰ Membership peaked at roughly 75 firms in 2016 before declining through delistings, suspensions, and qualifying exits, leaving about 30 firms by mid-2025.⁷¹ The firms that remained had therefore survived years of heightened exit risk. One later legal development altered the options open to a distressed issuer. Singapore enacted a court-supervised, Chapter 11-style restructuring regime in the Insolvency, Restructuring and Dissolution Act 2018, in force from 30 July 2020.⁷² A distressed issuer can therefore exit the cycle of watch-list inclusion, financing impairment, and depressed valuation through formal restructuring rather than delisting.⁷³

The watch-list also sat within a wider apparatus of post-listing instruments administered by SGX RegCo: public queries issued when unusual trading was detected, trade-with-caution alerts,

⁶⁷Bo Becker, Rüdiger Fahlenbrach and Ehsan Mahdikhani, ‘The Global Rise of Index-Based Ownership of Firms’ (Working Paper).

⁶⁸There was no equivalent watch-list for SESDAQ, later Catalist, issuers, which were instead monitored by their sponsors.

⁶⁹Whereas Nasdaq’s continuing-listing framework turns substantially on a minimum bid price, Singapore removed a comparable minimum-trading-price requirement in 2020, leaving the watch-list focused on loss-making companies with a market capitalisation below S\$40 million.

⁷⁰‘Singapore Exchange to Delist 5 Companies under Watch-List’, *Singapore Business Review*, <https://sbr.com.sg/financial-services/news/singapore-exchange-delist-5-companies-under-watch-list>.

⁷¹Our equity-only regression sample contains five firms with active watch-list status at the October 2025 removal event (E4), the difference from the full list reflecting firms suspended or delisted before the event window or excluded as non-equity instruments. Only these five had at least 120 valid trading days in the estimation window for cumulative abnormal returns around E4. The five are A31, AZA, FQ7, QS9, and R14; a sixth watch-listed firm, 5GI, was delisted before the event date.

⁷²Insolvency, Restructuring and Dissolution Act 2018 (No 40 of 2018), in force 30 July 2020

⁷³Meng Seng Wee and Hans Tjio, ‘Singapore as International Debt Restructuring Centre: Aspiration and Challenges’ (2022) 57 *Tex Intl LJ* 1

and trading suspensions for issuers with going-concern problems. Each intervention is publicly disclosed as it occurs. Together with watch-list status, these interventions form the cumulative public record of regulatory attention which our empirical analysis treats as the firm-level measure of the ‘information channel’. These instruments carry particular weight in a landscape where public enforcement dominates and private substitutes are scarce.⁷⁴ Recognising this scarcity, the reform package advanced investor-recourse proposals, later specified to operate through an independent designated representative coordinating action for affected investors.⁷⁵

3.4 The Equities Market Review Group

On August 2, 2024, MAS announced the formation of the Equities Market Review Group. It was chaired by Mr Chee Hong Tat, then Minister for Transport, Second Minister for Finance, and Deputy Chairman of MAS, and it was made up of senior public- and private-sector members. The Review Group was charged with recommending measures to strengthen the competitiveness of Singapore’s equities market within a twelve-month horizon, and it published its first set of measures in February 2025 and its final report in November 2025. The Review Group’s output proceeded through two workstreams. Each maps onto one of the reform’s principal instrument types and one of the two pricing channels examined below. First, the Enterprise and Markets workstream addressed issuer and investor participation, liquidity, and valuation quality through capital deployment, most prominently the S\$5 billion EQDP that MAS committed for deployment through professional fund managers investing in SGX-listed equities. Second, the Regulatory workstream covered the listing process, the disclosure-based regime, corporate-governance standards, and investor recourse; its principal instrument was SGX RegCo’s loosening of post-listing oversight. As a matter of legal authority, the first set of instruments rests on MAS’s developmental mandate and capital commitment,⁷⁶ while the second set was issued by SGX RegCo under its delegated listing-regulation authority.⁷⁷ Alongside these instruments, the package advanced listing-rule amendments that reduced the Mainboard profit threshold and streamlined qualitative admission and prospectus requirements.

⁷⁴Wai Yee Wan, Christopher Chen and Say H Goo, ‘Public and Private Enforcement of Corporate and Securities Laws: An Empirical Comparison of Hong Kong and Singapore’ (2019) 20 EBOR 319

⁷⁵Monetary Authority of Singapore, *Consultation Paper on Measures to Enhance Investor Recourse Avenues in Market Misconduct Cases* (24 October 2025).

⁷⁶The EQDP capital is drawn from MAS’s investment portfolio and the Financial Sector Development Fund: MAS, February 2025 announcement, para 3.5. For deployment from the investment portfolio, MAS Act s 4(2)(d), read with s 13B (power to appoint any person to perform MAS’s functions) and s 23(1)(m) (power to purchase and sell securities and investments approved by the board); for the Fund, MAS Act s 128(1)(a) (the Fund may be applied towards promoting Singapore as a financial centre), read with s 129(1) (power to invest the monies of the Fund) and s 129(2) (power to delegate that investment power to an agent).

⁷⁷Securities and Futures Act 2001 (2020 Rev Ed) s 320(1) (empowering MAS to delegate its supervisory functions).

3.5 Timeline of Reform Events

As shown in Table 1, the reform unfolded over sixteen months through a sequence of discrete announcements. The first substantive announcement (E1) bundled both instrument types, while later events activated narrower and partially overlapping subsets. For our purposes, this progressive unbundling generates the cross-event variation in policy content that we exploit.

The formation announcement on August 2, 2024 (E0) set the policy direction without specific measures beyond the twelve-month reporting commitment. We exclude it from hypothesis testing because its window overlaps the JPY carry-trade unwind of late July and early August 2024.⁷⁸ The first substantive announcement, on February 21, 2025 (E1), set out the Review Group’s measures. On the demand side, these included the S\$5 billion EQDP committed by MAS, fund-manager tax exemptions, an adjustment to Option C of the Global Investor Programme, and an expansion of the Grant for Equity Market Singapore (GEMS) research-grant scheme, alongside cross-border depository-receipt collaborations.⁷⁹ On the supply side, the regulatory component proposed a shift towards a disclosure-based regime through consolidation of listing review under SGX RegCo, together with streamlined qualitative admission and prospectus requirements; the remaining measures addressed investor recourse, shareholder engagement, and trading and settlement efficiency.

Subsequent events unbundled the February package along two parallel tracks. On the supervisory track, which activates the information channel, SGX RegCo issued its consultation paper at E2 (May 15, 2025), outlining a more disclosure-based regime. It proposed removing the financial watch-list, a reduced Mainboard profit threshold, a shift from public queries to private engagement, time-limited trade-with-caution alerts, and narrowed suspension criteria.⁸⁰ SGX RegCo then put these changes into effect at E4 (October 29, 2025), abolishing the watch-list and replacing public trading queries with private engagement when unusual trading is detected.⁸¹ On the developmental track, which activates the capital channel, MAS appointed the first three EQDP asset managers (Avanda Investment Management, Fullerton Fund Management, and J.P. Morgan Asset Management) at E3 (July 21, 2025). It placed S\$1.1 billion in initial capital with those managers and disclosed mandate terms specifying significant allocation to small- and mid-cap equities. MAS also committed S\$50 million to equity research and the listed-product ecosystem,

⁷⁸Each event window was screened for confounding news by reviewing MAS and SGX announcements, Singapore macroeconomic releases, exchange-wide trading notices, and major regional equity-market developments within the $(-1, +3)$ window. For E1 through E5 the screen identified no concurrent Singapore or regional equity-market dislocation capable of explaining the reported cross-sectional patterns; E0 is the sole exception, for the reason given in the text.

⁷⁹Option C of the Global Investor Programme, the family-office route, requires a Singapore-based single family office with at least S\$200 million under management; from 21 February 2025 new applicants must deploy at least S\$50 million into equities listed on approved Singapore exchanges. GEMS co-funds equity-research coverage of SGX-listed issuers.

⁸⁰Singapore Exchange Regulation, ‘SGX RegCo Consults on Measures in Line with a More Disclosure-Based Regime’, news release, 15 May 2025.

⁸¹Singapore Exchange Regulation, ‘SGX RegCo Advances Disclosure-Based Regime; Proposes Rule Changes on MAS’ Proposal to Consolidate Listing Review Functions under SGX RegCo’, news release, 29 October 2025.

and set out proposals to enhance investor recourse.⁸² At E5 (November 19, 2025), the Review Group published its final report together with the SGX–Nasdaq dual-listing bridge, which permits a primary listing on both boards using a single set of offering documents. It also announced a second batch of six EQDP managers with mandates totalling S\$2.85 billion. The accompanying market-structure measures — a board-lot reduction and designated-market-maker incentives — operate outside the two primary channels.⁸³ The dual-listing bridge raises a selection problem that our design cannot resolve. Easier access to a foreign market may induce additional cross-listings, including involuntary cross-listings whose valuation effects are mixed,⁸⁴ while weaker firms may be especially likely to select into the dual-listing regime.

4 Empirical Approach

We begin from a simple empirical premise: returns around each reform announcement provide evidence of how the market priced the reform’s two channels across firms. We apply that premise by measuring announcement returns for the same set of SGX-listed equities at each event in the reform timeline and comparing those returns across firms (Table 1).⁸⁵ The remainder of this section describes the data, specifies the estimation, and identifies the limits of what our design can establish.

Daily end-of-day prices for SGX-listed securities are drawn from SGX’s proprietary data services division for August 2022 through August 2025. Because the data licence ended in August 2025, the two later events (the October watch-list abolition (E4) and the November final report (E5)) use Yahoo Finance daily closing prices. A twenty-trading-day overlap validates that substitution against the SGX series, with a median price correlation of essentially one.⁸⁶

⁸²Monetary Authority of Singapore, ‘MAS Appoints First Batch of EQDP Asset Managers; Commits S\$50 Million to Boost Equity Research and Product Listings; and Outlines Proposals to Enhance Investor Recourse’, media release, 21 July 2025.

⁸³Monetary Authority of Singapore, ‘Review Group Completes Equities Market Review, Unveils SGX–Nasdaq Dual Listing Bridge, S\$30 Million “Value Unlock” Package, and Second Batch of EQDP Asset Managers’, media release, 19 November 2025.

⁸⁴Peter Iliev, Darius P Miller and Lukas Roth, ‘Uninvited U.S. Investors? Economic Consequences of Involuntary Cross-Listings’ (2014) 52 J Acct Res 473

⁸⁵The use of security-price reactions to measure the effects of regulation follows G William Schwert, ‘Using Financial Data to Measure Effects of Regulation’ (1981) 24 J L & Econ 121, and the event-study methodology surveyed in A Craig MacKinlay, ‘Event Studies in Economics and Finance’ (1997) 35 J Econ Lit 13; the same cross-sectional approach is applied to a corporate-law reform in Kenneth Khoo and Roberto Tallarita, ‘The Price of Delaware Corporate Law Reform’ (SSRN Working Paper No 5318203, 2025)

⁸⁶The SGX price data, monthly listings snapshots, issuer feeds, and corporate-announcement records used here are proprietary and were obtained under a data licence from SGX covering August 2022 to August 2025. Yahoo Finance prices, used only for E4 and E5, do not carry bid–ask quotes, which precludes post-event market-quality analysis for those two events but does not affect the abnormal-return estimates. For these two events the SGX equal-weighted index used as the CAPM benchmark is likewise recomputed on the intersection of equities with valid Yahoo observations (≈ 716 stock codes), so the composition shift affects only the CAPM market return there; the externally sourced Fama–French factors are unaffected.

For abnormal returns, the benchmark is the Fama–French three-factor series for the Asia Pacific ex Japan region,⁸⁷ taken from the Dartmouth data library. Fama and French show that regional factors capture international return premia more accurately than global factors; on that basis, we use the regional benchmark.⁸⁸ Firm characteristics come from SGX monthly market-capitalisation snapshots, annual financial-statement data,⁸⁹ SGX regulatory actions parsed from corporate-announcement feeds, and board and sector information from SGX issuer registries.

The sample is confined to ordinary equity securities. It excludes warrants, structured products, daily leverage certificates, exchange-traded funds, real estate investment trusts, and business trusts, thereby preserving the population of operating firms to which the reform was addressed. Each firm’s factor loadings (its sensitivities to the benchmark factors) are estimated over a fixed pre-E1 window, from January 2, 2024 to February 14, 2025. Because every event uses that same window, the reform announcements themselves cannot contaminate the loadings. Each firm must have at least 120 valid return observations in that window. The requirement improves the precision of the loadings on which abnormal returns depend, although it also removes the most thinly traded counters, which are concentrated in the small-cap and Catalist tiers where the channel effects are hypothesised to be strongest. The resulting exclusions tend to bias the estimated effects towards zero.⁹⁰ Together, the equity filter and the trading requirement leave roughly 260 to 275 firms with valid abnormal returns at each event.

Firm size proxies the capital channel, while the regulatory friction count proxies the information channel. Our primary size measure is an above-median indicator, equal to one when a firm’s log market capitalisation exceeds the event-specific sample median. Because the size distribution is heavily right-skewed and investability tends to operate as a threshold, we treat the indicator as primary and retain continuous log market capitalisation as a secondary measure. Both measures are taken from the most recent monthly snapshot before the event window. The friction count records the cumulative number of SGX queries, trading halts, and trading suspensions a firm had attracted as of the event date, together with rarer formal regulatory actions. We later sharpen this measure in two ways: first, by separating routine queries from the more consequential halts and suspensions; and second, by separating friction accumulated in the preceding twelve months from older friction.⁹¹ A watch-list indicator marks the five firms carrying SGX’s Financial

⁸⁷Eugene F Fama and Kenneth R French, ‘Common Risk Factors in the Returns on Stocks and Bonds’ (1993) 33 J Fin Econ 3

⁸⁸Eugene F Fama and Kenneth R French, ‘Size, value, and momentum in international stock returns’ (2012) 105 J Fin Econ 457

⁸⁹Annual financial-statement data are drawn primarily from Worldscope via WRDS and supplemented by figures extracted from approximately 17,000 SGX filing PDFs through a three-pass procedure combining automated table extraction, a vision-language model for unstructured layouts, and manual review flags for approximately 70 rows with known scale or currency ambiguities.

⁹⁰Among firms excluded for insufficient trading, the median count of valid return days is fewer than sixty against a calendar of roughly 280 trading days.

⁹¹The friction count is built from eight SGX “Announcement Category” classes, matched exactly on the structured CSV feed: (1) Response to SGX Queries, (2) Query Regarding Trading Activity, (3) Request for Trading Halt, (4) Request for Lifting of Trading Halt, (5) Request for Suspension, (6) Request for Resumption of Trading from Suspension, (7) Regulatory Actions by SGX, and (8) Announcement in Relation to Regulatory Actions by SGX and/or Other Authorities. Friction accumulates over August 2022 to August 2025 for E0–E3 and is extended through the event date for E4 and E5. The type decomposition groups (1)–(2) as queries, (3)–(6) as halts and suspensions, and (7)–(8)

Watch-List status at the October abolition event. The regressions also control for illiquidity (the standardised Amihud measure of how strongly trading moves a firm’s price),⁹² Catalyst listing status,⁹³ accounting characteristics, and sector indicators so that firms are compared within two-digit industry sectors.

For each event, a firm’s abnormal return is the difference between its actual daily return and the return the benchmark predicts. The benchmark regression relates each firm’s daily return to three region-wide factors: the overall market return (MKT), the return difference between small and large firms (SMB), and the return difference between value and growth firms (HML). Asterisks mark returns adjusted for thin trading, and w_t is the adjusted intercept term that replaces the usual constant; both adjustments are described below. The benchmark is then

$$R_{i,t}^* = \hat{\alpha}_i w_t + \hat{\beta}_{1,i} \text{MKT}_t^* + \hat{\beta}_{2,i} \text{SMB}_t^* + \hat{\beta}_{3,i} \text{HML}_t^* + \hat{\varepsilon}_{i,t}, \quad (1)$$

and the residuals — the daily differences between actual and predicted returns — are cumulated over the five-day event window, from one trading day before to three trading days after the announcement, to give the firm’s cumulative abnormal return.⁹⁴ Following Maynes and Rumsey,⁹⁵ we apply a thin-trading correction to address the non-synchronous trading that is common on SGX, where many securities do not trade every day.⁹⁶ In the watch-list analysis, a secondary specification uses a single-factor market model benchmarked to an SGX equal-weighted index. We adopt it there because watch-list entry is defined partly by market capitalisation, so a benchmark that already adjusts returns for firm size would remove much of the variation the analysis needs to detect.⁹⁷ The descriptive statistics for the sample, together with the firm-size distribution that motivates the median split, appear in [Table 2](#) and [Figure 1](#).

At the cross-sectional stage, we relate these abnormal returns to the size and friction proxies by estimating the same specifications at every event. The specifications build up in steps across the main tables, so that we can track the stability of each estimate as controls enter. First, the variable of interest enters on its own, establishing the unconditional association. Second, the other channel’s variable is added, testing whether the two proxies absorb one another or whether they instead reflect a single underlying characteristic. Third, controls enter for firm financials, market microstructure (including liquidity and trading frictions), listing tier, and industry sector. Where we pool adjacent events, as in the February-to-July reversal, each firm appears at both events, so

as regulatory actions. Halts and suspensions, although frequently issuer-initiated under the exchange’s disclosure regime, mark publicly observable trading interruptions that typically accompany scrutiny or disclosure problems; the measure therefore captures the public record of scrutiny-related events rather than regulator-initiated actions alone.

⁹²Yakov Amihud, ‘Illiquidity and stock returns: cross-section and time-series effects’ (2002) 5 J Fin Mkts 31

⁹³The Catalyst board-tier effect is an artefact of sample composition and is absorbed by firm size; see [Table 22](#).

⁹⁴Estimates under alternative event windows are reported in [Table 17](#).

⁹⁵Elizabeth Maynes and John Rumsey, ‘Conducting event studies with thinly traded stocks’ (1993) 17 J Banking & Fin 145

⁹⁶When a firm does not trade for d_t consecutive days, the procedure cumulates the firm’s return and the factor returns over the non-trading gap and divides all variables by $\sqrt{d_t}$ to correct for the heteroskedasticity induced by the unequal return horizons. It omits the conventional constant and instead includes a thin-trading-adjusted intercept regressor, $w_t = 1/\sqrt{d_t}$, because in a thin market a conventional intercept could absorb genuine abnormal returns into the alpha estimate when many zero-return days are present.

⁹⁷The core size and friction results are robust across the alternative factor models reported in [Table 18](#).

we adjust the standard errors for the correlation within each firm’s pair of observations (clustering by firm). To limit the influence of outliers, we winsorise abnormal returns, resetting values beyond the 1st and 99th percentiles to those percentiles. For single-event regressions, we use standard errors that remain valid when return variability differs across firms (heteroskedasticity-robust, HC1). As the abnormal returns are themselves estimates from the benchmark regression, we corroborate the headline coefficients with a bootstrap that carries that estimation error into the second stage (Section A.2.6).⁹⁸

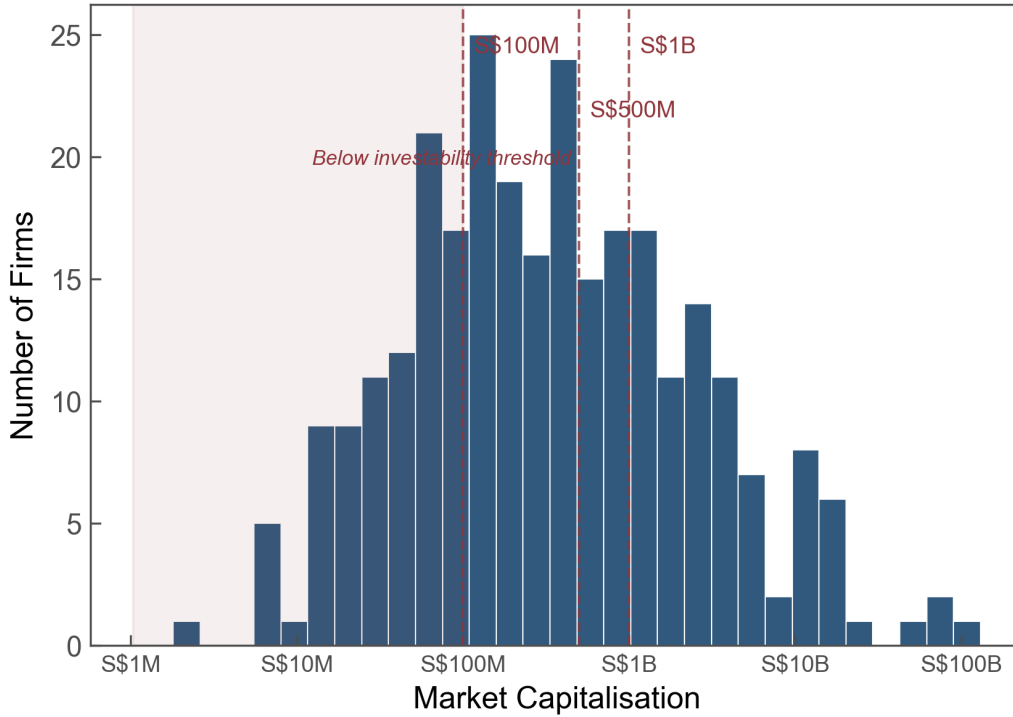
Table 2: Descriptive Statistics (E1 Sample)

	<i>N</i>	Mean	SD	Min	Median	Max
CAR(−1, +3)	274	-0.013	0.055	-0.195	-0.006	0.115
log(Market cap)	283	19.637	1.958	15.827	19.501	24.774
Friction count	285	2.475	3.302	0.000	1.000	17.160
Watchlist	285	0.018	0.132	0.000	0.000	1.000
Above median	285	0.498	0.501	0.000	0.000	1.000
ROA (%)	257	1.848	14.242	-78.143	3.855	19.459
Leverage	258	23.874	18.016	0.000	22.869	67.866
Book-to-market	249	1.540	1.111	0.032	1.325	5.034
Illiquidity (z-Amihud)	283	-0.176	0.159	-0.217	-0.217	0.749
Catalist	285	0.147	0.355	0.000	0.000	1.000

Notes: Sample comprises 285 SGX-listed equities in the E1 cross-section (2025-02-21); 274 have valid FF3 CARs. Market cap is from the last available trading day before the event window. Illiquidity (z-Amihud) is the standardised Amihud illiquidity measure over the 60 trading days before the event. Friction count is the cumulative number of SGX queries, trading halts, and trading suspensions accumulated by the firm as of the event date; watch-list is reported as an indicator variable. Broker ownership is the share of top-20 holdings registered through retail securities brokers; available for 205 firms with shareholder register data. Accounting variables (ROA, Leverage) are available for firms with Worldscope coverage ($N = 257$ – 258). Catalist and watch-list are indicator variables.

⁹⁸Adrian Pagan, ‘Econometric Issues in the Analysis of Regressions with Generated Regressors’ (1984) 25 Intl Econ Rev 221

Figure 1: Distribution of market capitalisation across the SGX equity-only sample



Note: The figure illustrates the distribution of market capitalisation across the SGX equity-only sample, shown on a log-scaled x-axis. Market capitalisation is from the last available SGX monthly snapshot before E1. Dashed vertical lines mark illustrative investability thresholds at S\$100 million, S\$500 million, and S\$1 billion; the shaded region highlights firms below the S\$100 million threshold.

These specifications identify how the market priced the size and friction proxies across firms, but they do not recover the total value assigned to either policy instrument. Our design relies principally on a cross-sectional decomposition of the bundled February announcement (E1), where both proxies are priced at the same event. Subsequent events provide only partial separation of the two channels. The May consultation (E2) generated no average market response, perhaps because the principal proposals had already been signalled, while the July appointment of asset managers (E3) reversed the size gradient without cleanly isolating the effect of capital deployment. The November event (E5) likewise bundled market-structure reforms with further capital deployment and the dual-listing bridge, preventing separate identification of the market-structure component.

The capital channel turns on whether a firm is investable to professional managers, a characteristic with no single direct measure. Its observable markers — firm size, analyst coverage, and foreign institutional ownership — are strongly collinear, because large firms also tend to attract analyst coverage and foreign institutions. A regression that included all three could not tell their effects apart, so we use firm size as the single proxy for investability and treat the coverage and ownership evidence in [Section 5.1](#) as corroboration. Because only five firms are treated, the watch-list result provides suggestive evidence of stigma relief and is necessarily imprecise. More generally, our findings are specific to the SGX setting and may not extend to markets with different investor

bases, regulatory structures, or levels of secondary-market liquidity (Section 6).

5 Findings

We organise the findings around the two channels and trace how each behaves across the reform sequence, beginning with the average market response at each event. Table 3 reports the average abnormal return at each event. The bundled February announcement (E1) produced a mean cumulative abnormal return of -1.32 percent, significant at the 1% level, and only 40.5% of firms posted positive returns, so the typical firm lost value on the news. The May consultation (E2) drew a null response (-0.11 percent, $p = 0.721$), consistent with its having been anticipated once E1 signalled that detailed proposals would follow. Each of the later events was positive and significant at the 1% level. In order, the July manager appointment (E3) produced a mean of 3.12 percent with 70.0% of firms gaining, the October watch-list abolition (E4) a mean of 1.20 percent, and the November final report (E5) a mean of 0.62 percent. The shift from a negative average at E1 to a positive average at E3 motivates the analysis across firms that follows, because the aggregate figures hide the differentiation by size and friction that the channels predict.⁹⁹

Table 3: Event-Level Summary Statistics

Event	Date	Channel	N	Mean CAR	t-stat	p-value	Median	% Pos
E0	2024-08-02	Signal	260	-0.0215^{***} (0.0041)	-5.25	0.000	-0.0187	30.8%
E1	2025-02-21	Capital + Information	274	-0.0132^{***} (0.0033)	-3.95	0.000	-0.0058	40.5%
E2	2025-05-15	Information	275	-0.0011 (0.0031)	-0.36	0.721	-0.0037	42.9%
E3	2025-07-21	Capital	270	0.0312^{***} (0.0046)	6.82	0.000	0.0199	70.0%
E4	2025-10-29	Information (implementation)	258	0.0120^{***} (0.0040)	2.99	0.003	0.0035	55.8%
E5	2025-11-19	Capital	258	0.0062^{***} (0.0022)	2.76	0.006	0.0063	60.5%

Notes: The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses; significance is assessed by the Student t -test. % Pos is the percentage of firms with positive CARs at that event. “Channel” indicates which of the article’s two main channels—the capital channel (deployment through fund managers) and the information channel (deregulation of SGX’s oversight regime)—each event primarily activates. E5 also includes market structure enhancements (board lot reduction, market maker incentives) that operate outside the two primary channels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ for H_0 : mean = 0.

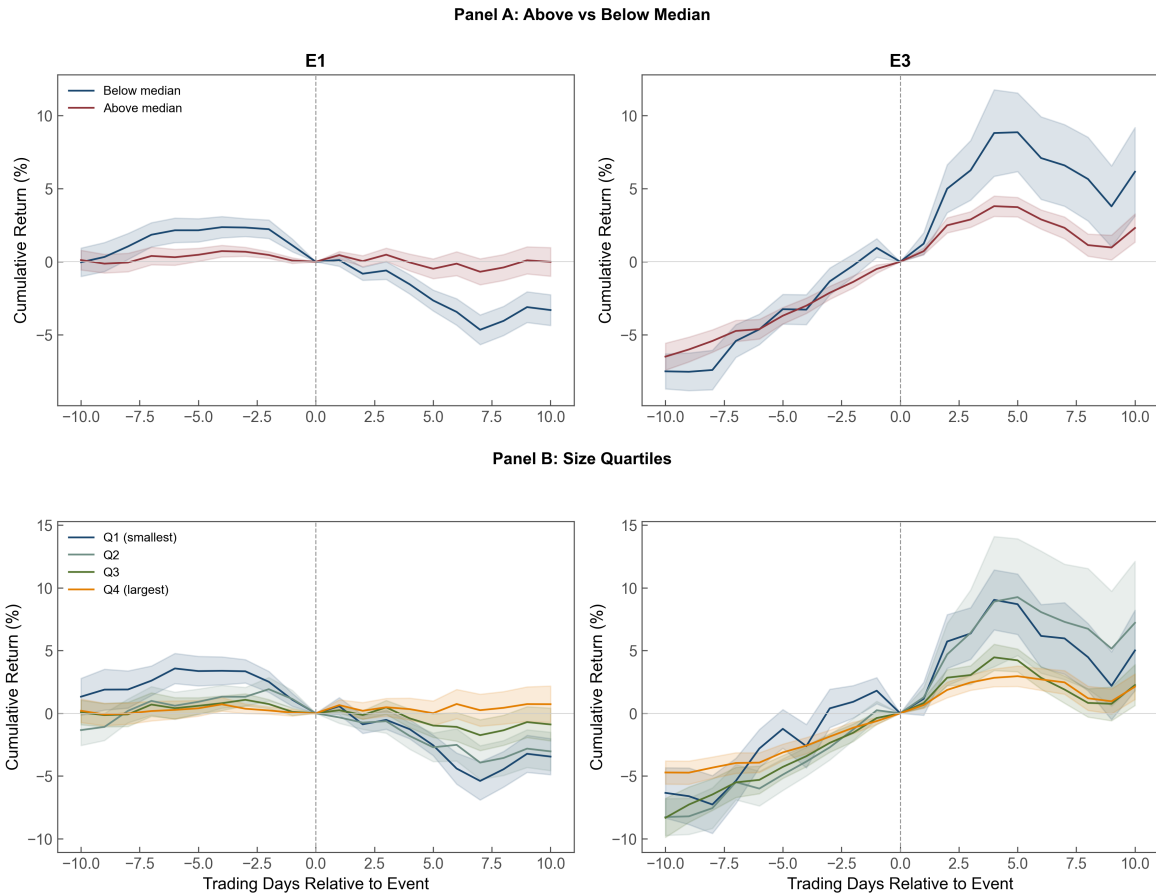
⁹⁹Event-level cross-sectional detail for E0, E2 and E3, and E5 is reported in Sections A.4.1 to A.4.3.

5.1 The Intermediation Gap and the Mandate-Specificity Reversal

A S\$5 billion programme presented as support for small- and mid-cap equities would ordinarily be expected to benefit those firms directly. The announcement returns point in the opposite direction: the market priced the EQDP as a large-cap event. As discussed in [Section 2.1](#), the EQDP committed the capital through professional managers whose investability constraints favour large and liquid securities,¹⁰⁰ so larger firms should have posted higher announcement returns if the market anticipated those constraints. Sorting firms at E1 into market-capitalisation quartiles, we find that the smallest quartile recorded a mean cumulative abnormal return of −2.54 percent, significant at the 1% level, whereas the largest quartile gained 0.22 percent, which is statistically indistinguishable from zero. The spread between the extreme quartiles was therefore 2.76 percentage points, significant at the 1% level ([Figure 3](#)). Moreover, the gradient rises step by step across all four quartiles. This increasing pattern is what a binding investability constraint predicts. Indeed, a reaction specific to the very smallest firms would instead have concentrated the losses in the bottom quartile. [Table 4](#) reports the quartile and median-split means, while [Figure 2](#) plots the daily cumulative abnormal-return paths that trace the E1 decline and the E3 reversal directly.

¹⁰⁰On the fiduciary-duty component of those constraints, see Law Commission, *Fiduciary Duties of Investment Intermediaries* (Law Com No 350, 2014)

Figure 2: Cumulative abnormal return paths for above- and below-median firms across E1 and E3



Note: Panel A: Daily cumulative log returns for firms above (navy) vs. below (maroon) median log market capitalisation, normalised to zero at day 0. Panel B: Daily cumulative log returns by size quartile (Q1 = smallest). Left column: E1 (February 21, 2025). Right column: E3 (July 21, 2025). The E1-to-E3 reversal in the size gradient is visible in both panels.

Table 4: Mean CARs by Size Quartile and Median Split (E1 and E3)

Panel A: By market-capitalisation quartile

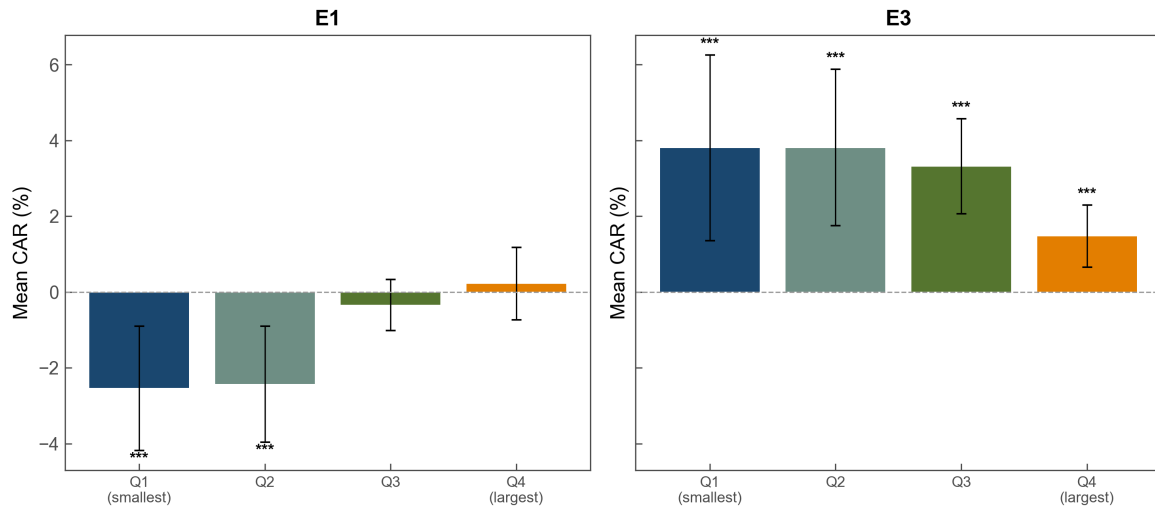
Quartile	E1	E3
Q1 (smallest)	-2.54 ^{***} (0.85)	+3.83 ^{***} (1.27)
Q2	-2.43 ^{***} (0.79)	+3.83 ^{***} (1.07)
Q3	-0.34 (0.34)	+3.32 ^{***} (0.64)
Q4 (largest)	+0.22 (0.49)	+1.48 ^{***} (0.42)
$Q4 - Q1$	+2.76 ^{***} (2.83)	-2.36 [*] (-1.76)
N	272	269

Panel B: By above/below-median split

	E1	E3
Below median	-2.57 ^{***} (0.58)	+4.10 ^{***} (0.88)
Above median	-0.08 (0.29)	+2.34 ^{***} (0.42)
Difference	+2.49 ^{***} (3.82)	-1.75 [*] (-1.79)
N	274	270

Notes: Mean cumulative abnormal returns (%) over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Panel A: size quartiles formed on log market capitalisation within each event. Panel B: groups formed on above/below median log market capitalisation within each event. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for quartile and group means; significance is assessed by the Student t -test. Welch t -statistics in parentheses for the $Q4 - Q1$ and two-sample differences. ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

Figure 3: Mean cumulative abnormal returns by market capitalisation quartile across E1 and E3



Note: Mean FF3 CAR($-1, +3$) by market capitalisation quartile. Error bars show 90% confidence intervals. At E1, returns rise steadily with size ($Q4 - Q1 = 2.76$ pp, $p < 0.01$). At E3, the gradient reverses, with smaller firms gaining more ($Q1 = 3.83\%$, $Q4 = 1.48\%$), consistent with the mandate terms targeting small- and mid-cap allocation. The stacked E1–E3 interaction estimates a change in the above-median effect of $\gamma_2 = -0.042$, significant at the 1% level (Table 6), and a paired-date placebo places the observed reversal at the 2nd percentile of its null distribution ($p = 0.030$; Table 14).

The firm-level results confirm the size gradient. In the median-split comparison, firms above the median of log market capitalisation recorded mean abnormal returns of -0.08 percent at E1, while firms below that threshold recorded -2.57 percent; the resulting 2.49 percentage-point spread is statistically significant at the 1% level (Table 4). The regression specification points in the same direction: the above-median size indicator carries a coefficient of roughly 2.3 percentage points, significant at the 1% level, and remains stable as financial controls and the friction and watch-list variables are added in turn (Table 5). When size is measured continuously, each log-unit of market capitalisation is associated with announcement returns 0.54 percentage points higher — a doubling of firm size corresponds to roughly 0.4 percentage points — again significant at the 1% level (Tables 19 and 20). This evidence is consistent with the prediction that capital routed through professional managers would favour firms already inside the investable universe.

Table 5: Size and Announcement Returns: Regression Specifications (E1 and E3)

	E1					E3				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Above median	0.0249*** (0.0065)	0.0227*** (0.0062)	0.0211*** (0.0071)	0.0175*** (0.0067)	0.0203** (0.0096)	-0.0175* (0.0098)	-0.0190** (0.0094)	-0.0148 (0.0102)	-0.0127 (0.0101)	0.0025 (0.0117)
Friction count		-0.0031** (0.0012)	-0.0034** (0.0016)	-0.0031** (0.0012)	-0.0032** (0.0016)		0.0004 (0.0016)	0.0013 (0.0019)	0.0009 (0.0016)	0.0015 (0.0015)
Watchlist		0.0524* (0.0316)	0.0575 (0.0360)	0.0512 (0.0327)	0.0559 (0.0370)		-0.0612 (0.0697)	-0.0763 (0.0744)	-0.0126 (0.0692)	0.0100 (0.0655)
Illiquidity				-0.0018 (0.0098)	-0.0014 (0.0110)				-0.0203* (0.0106)	-0.0253*** (0.0090)
Catalist				-0.0098 (0.0126)	-0.0001 (0.0176)				0.0199 (0.0192)	0.0200 (0.0245)
Financial controls	No	No	Yes	No	No	No	No	Yes	No	No
Sector FE	No	No	No	No	Yes	No	No	No	No	Yes
<i>N</i>	274	274	240	272	243	270	270	238	268	248
<i>R</i> ²	0.051	0.099	0.135	0.099	0.205	0.013	0.023	0.051	0.043	0.246

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Columns (1)–(5): E1; Columns (6)–(10): E3. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median within each event. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Once the mandate terms became specific, the relationship between size and returns reversed. At the July appointment of the first three EQDP managers (whose mandate terms specified significant allocation to small- and mid-cap equities), the smallest quartile gained 3.83 percent, while the largest gained 1.48 percent; both estimates are significant at the 1% level. In [Figure 3](#), this difference appears as a negative and marginally significant size spread of -2.36 percentage points. The regression evidence confirms the reversal. The median split that favoured above-median firms at E1 reverses at E3, where below-median firms hold a 1.75 percentage point advantage. As shown in [Table 5](#), the above-median coefficient reaches -1.9 percentage points at the 5% level at E3, while the friction count has no explanatory power there. Pooling E1 and E3 in a stacked design, in which each firm is observed at both events, yields the same pattern — above-median firms posted returns 2.5 percentage points higher at E1, but their relative performance shifted by 4.2 percentage points in the opposite direction at E3; both estimates are significant at the 1% level ([Table 6](#)). The estimated swing stays between 3.8 and 4.7 percentage points, and remains significant at the 1% level, as financial controls, the friction count, and sector effects are added in turn, so the reversal does not turn on any particular set of controls.

Table 6: Stacked E1–E3 Size Interaction

	(1)	(2)	(3)	(4)
Above median	0.0249 ^{***} (0.0065)	0.0280 ^{***} (0.0078)	0.0232 ^{***} (0.0071)	0.0264 ^{***} (0.0092)
Post-E3	0.0667 ^{***} (0.0105)	0.0699 ^{***} (0.0111)	0.0557 ^{***} (0.0112)	0.0542 ^{***} (0.0114)
Above median $\times$ Post-E3	-0.0424 ^{***} (0.0116)	-0.0467 ^{***} (0.0122)	-0.0397 ^{***} (0.0110)	-0.0377 ^{***} (0.0113)
Friction count			-0.0033 ^{**} (0.0016)	-0.0036 [*] (0.0021)
Friction $\times$ Post-E3			0.0044 (0.0032)	0.0045 (0.0035)
Financial controls	No	Yes	Yes	Yes
Sector FE	No	No	No	Yes
N	544	478	478	471
R^2	0.126	0.142	0.155	0.249

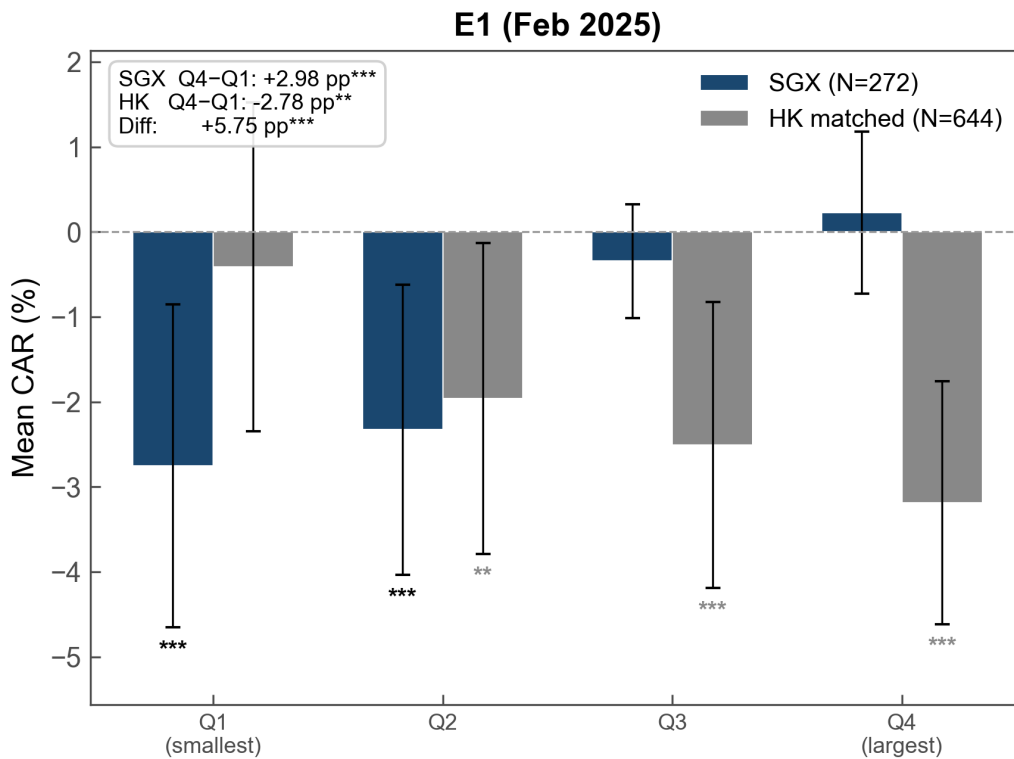
Notes: OLS estimates of the stacked E1–E3 size specification with event fixed effects (captured by Post-E3 = 1 for E3 observations) and standard errors clustered by stock code, because each firm contributes two observations (one at E1, one at E3); HC1 would understate the standard errors by treating these as independent, when in reality the same firm’s CARs at E1 and E3 are correlated. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Column (1) reports the bare stacked specification. Column (2) adds financial controls (ROA, leverage, and book-to-market). Column (3) further adds the friction count and its E3 interaction as a robustness check; this column corresponds to column (3) of [Table 11](#). Column (4) augments column (3) with two-digit SSIC sector fixed effects. ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

Much of our empirical case rests on this reversal. To assess whether a movement of this size could arise by chance, we use two placebo comparisons. The first is a ‘paired-date placebo’: two hundred pairs of pre-reform pseudo-event dates, each pair separated by the same interval as E1 and E3, re-estimated under the identical specification ([Section A.1.1](#) details the construction). The observed reversal is extreme against this benchmark: only six of the two hundred placebo pairs produce a swing of comparable magnitude ($p = 0.030$), and the reversal in the continuous size measure is similarly unusual ([Table 14](#)). The placebo also rules out two competing readings of the sequence. First, small-cap SGX equities might have been oversold before E3, so the subsequent movement could reflect mean reversion on positive news rather than repricing mandate credibility. Because the paired-date design folds any such tendency for small firms to bounce after declines into the null distribution, the observed reversal remains in the tail against that benchmark. Second, the pattern might reflect a generic small-cap rebound unrelated to the mandate. The event sequence is difficult to reconcile with that account: returns to small firms were negative at E1, null at E2,

and positive at E3 precisely when the mandate terms were disclosed. A generic rebound would not ordinarily be expected to coincide so closely with the disclosure of the mandate terms.

The remaining market-wide alternative attributes the pattern to a regional size rotation rather than a Singapore-specific treatment effect. We address that possibility with a ‘cross-market placebo’: each SGX firm is matched to its three nearest Hong Kong neighbours by market capitalisation, and the same specification is estimated on both markets at E1 ([Section A.1.2](#)). The two markets show opposite patterns ([Figure 4](#)): returns rise with size at every SGX quartile, whereas the matched Hong Kong sample shows a significantly negative size gradient, and the cross-market difference is significant at the 1% level ([Table 15](#)). A common regional shock to small-cap valuations should produce size gradients of the same sign across the two markets, rather than the opposing relationships observed here.

Figure 4: Cross-Market Placebo: Size Quartile CARs at E1 (SGX vs Matched HK)



Note: Bars show mean FF3 CARs (%) by size quartile at E1 (February 2025). Navy bars: SGX firms. Gray bars: HKEX firms matched by log market capitalisation ($K = 3$). Full regression results are available in [Table 15](#).

As shown in [Table 7](#), two further dimensions of investability move with size in the same way and corroborate the interpretation. Sorted by analyst coverage, firms with no coverage recorded mean abnormal returns of -1.80 percent at E1, significant at the 1% level, whereas firms followed by six or more analysts posted -0.42 percent, a return statistically indistinguishable from zero. At E3, where the GEMS research-grant expansion specifically targeted under-covered firms, this ordering reversed: zero-coverage firms gained 3.04 percent, compared with 1.05 percent for the

most covered firms (a rank correlation between coverage and returns of -0.18 , $p = 0.060$). Foreign institutional ownership traces the same E1 pattern, with top-quartile firms gaining 0.20 percent and bottom-quartile firms losing 2.36 percent (rank correlation 0.13, $p = 0.083$). Because each measure is strongly collinear with size, neither coverage nor foreign ownership retains significance once log market capitalisation is controlled. We therefore treat investability as a single composite proxied by firm size, using coverage and ownership only as corroborating evidence. This pattern accords with the mechanism identified by Didier,¹⁰¹ who observes that foreign institutional investors tend to concentrate in large firms because size proxies for the information environment that enables institutional participation.¹⁰²

¹⁰¹Tatiana Didier, 'Information Asymmetries and Institutional Investor Mandates' (Working Paper, 2011)

¹⁰²Retail-dominated firms, proxied by high broker ownership, display the same E1 discount and E1-to-E3 reversal; see [Tables 24 to 26](#).

Table 7: Analyst Coverage and Foreign Institutional Ownership

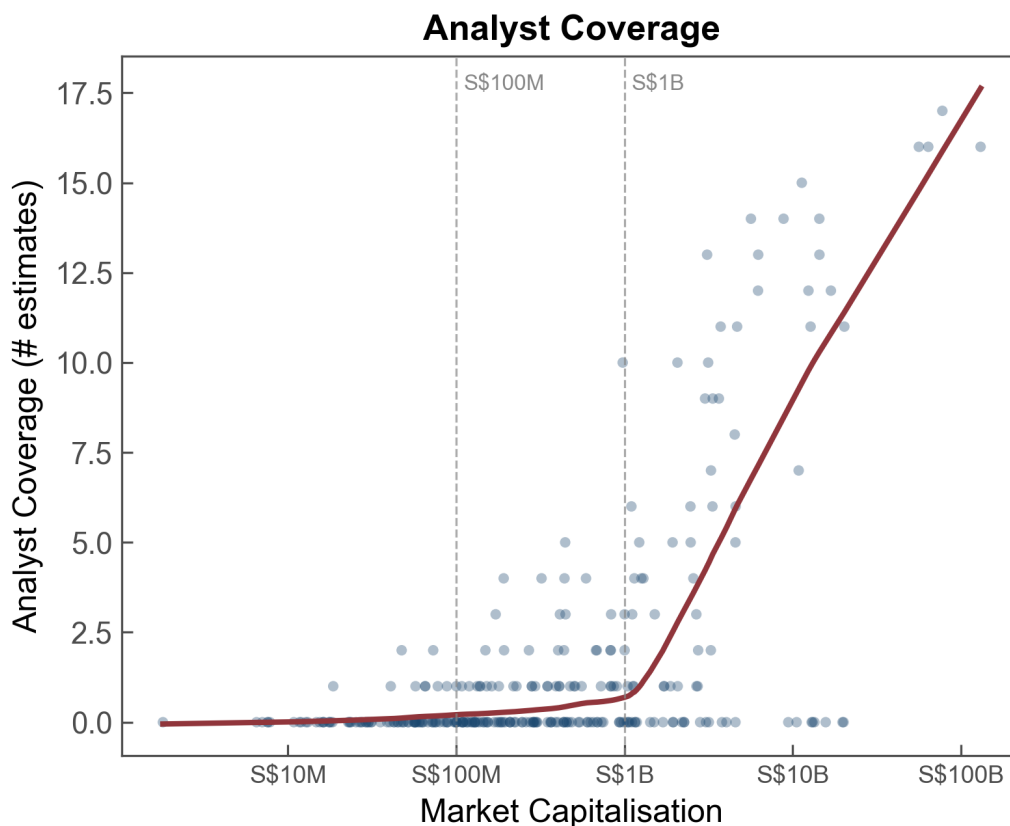
<i>Panel A: Mean CARs (%) by I/B/E/S Analyst Coverage Group</i>				
Coverage group	E1 (Feb 2025)		E3 (Jul 2025)	
	Mean CAR	N	Mean CAR	N
Zero coverage	−1.80 ^{***} (0.48)	165	+3.04 ^{***} (0.58)	163
1–2 analysts	−0.65 (0.81)	57	+2.81 ^{***} (0.69)	51
3–5 analysts	−0.23 (0.75)	21	+3.50 ^{**} (1.64)	22
6+ analysts	−0.42 (0.49)	31	+1.05 ^{**} (0.43)	34
Spearman ρ (covered only)	−0.083 (0.096)	109	−0.182 [*] (0.096)	107
<i>Panel B: Foreign Institutional Ownership Quartile CARs (E1 only)</i>				
	Mean CAR	Mean IO _{for}	N	
Q1 (lowest)	−2.36 ^{**} (0.96)	0.01%	46	
Q2	−0.67 (1.02)	0.23%	44	
Q3	−0.85 (0.70)	1.35%	45	
Q4 (highest)	+0.20 (0.68)	9.72%	46	
Spearman ρ	+0.129 [*] (0.074)		181	

Notes: Panel A: Mean CARs (%) by I/B/E/S analyst coverage group. Firms not in I/B/E/S are classified as zero coverage. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for group means; significance is assessed by the Student t -test. Asymptotic standard errors in parentheses for Spearman ρ (computed on covered firms only). Panel B: E1 only. Foreign institutional ownership (IO_{for}) from FactSet, most recent quarter $\leq$ event date (Q4 2024). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for quartile means; asymptotic standard error in parentheses for Spearman ρ . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The size dispersion of the SGX universe explains why the investability constraint operates as a threshold. With a total market capitalisation of just under S\$1 trillion, that universe runs from micro-cap firms valued at under S\$2 million to constituents exceeding S\$130 billion. Across this range, firms below roughly S\$100 million in capitalisation rarely achieve the institutional visibility or tradability needed to become investable, while firms above S\$1 billion generally clear both constraints. Figure 5 illustrates the discontinuity: sell-side coverage is near zero below approximately S\$1 billion before rising sharply above it. The binary above-median (market-capitalisation) indica-

tor is therefore more precise than its continuous measure, since it captures the transition from non-investable to investable status. In the United States, public capital channelled through fund managers can reach mid-sized firms because those firms already exceed the relevant visibility and liquidity thresholds; managers need only increase their portfolio weights. In Singapore, many of the programme's intended beneficiaries remain below those thresholds and outside the investable universe. This is the threshold dynamic predicted by the framework developed in [Section 2.1](#).

Figure 5: Analyst Coverage by Market Capitalisation



Note: Each point represents one SGX-listed firm. The I/B/E/S analyst estimate count rises sharply above approximately S\$1 billion in market capitalisation. Dashed lines at S\$100 million and S\$1 billion mark approximate thresholds.

Although the reversal narrowed the gap, it did not eliminate the advantage of larger firms. Cumulated across the February announcement and the July mandate disclosure, the net size effect still favoured above-median firms by approximately 0.7 percentage points.¹⁰³ The announcement reversal, its stability across specifications, and the two placebo comparisons together indicate a sequential pricing mechanism. The market initially priced the constraints faced by investment

¹⁰³A descriptive analysis of realised dollar volume across the reform sub-periods, reported in [Table 27](#) and [Figure 9](#) (see [Section A.4.6](#)), shows that, at the medians, the active-deployment window is the only post-E1 window in which volume grew proportionally faster for below-median firms. This ranking is descriptive only: the panel regression cannot distinguish the two groups within that window.

intermediaries, then revised that assessment once the disclosed mandate terms restricted the allocation of capital. This sequence is the intermediation gap expressed through announcement returns.

5.2 Deregulation as Information Loss

We turn to the deregulatory component of the February announcement, and with it to the information side of the intermediation gap. The question is how the market priced the proposed withdrawal of SGX's public oversight apparatus for the firms most exposed to it: those with the longest accumulated records of queries, halts, suspensions, and formal regulatory actions, which our friction count measures. As [Section 2](#) set out, the two accounts of securities regulation predict opposite reactions for these firms. If oversight is chiefly a compliance burden, the firms that attract the most regulatory attention bear the heaviest costs, so the prospect of lighter oversight should raise their prices relative to other firms. If oversight instead produces information that investors rely on to price firms whose quality they cannot easily verify, those same firms should fall, because they are the firms about which the market stands to lose the most. Whether high-friction firms gained or lost at E1 therefore reveals which account the market acted on.

The E1 cross-section indicates that the market priced firms' exposure to the information channel. In [Table 8](#), which relates each firm's E1 abnormal return to its friction count, firms with heavier regulatory histories lost more: each additional friction event predicts a decline of roughly one-third of a percentage point, significant at the 1% level in the univariate specification.¹⁰⁴ High-friction firms may simply be smaller, more thinly traded, or concentrated in troubled sectors, so the friction count could be standing in for some other characteristic the market was pricing. The remaining columns address that possibility by adding the size indicator, the watch-list indicator, financial controls, illiquidity, the Catalist indicator, and sector fixed effects in turn. If the estimated penalty merely reflected one of those characteristics, it should attenuate once the relevant control enters the regression. Instead, the estimate remains between 0.31 and 0.34 percentage points per friction event across the specifications.

The joint specification provides the strongest test. When the size indicator and friction count enter together, both remain significant and neither absorbs the other. Were they simply alternative measures of a common underlying characteristic, such as overall firm quality, the inclusion of both would be expected to weaken at least one coefficient substantially. Their independent explanatory power is consistent with the two-channel account: the market priced exposure to the capital programme through firm size and exposure to deregulation through the firm's regulatory history.¹⁰⁵

A final check concerns the benchmark itself. The three-factor model already adjusts each firm's return for its exposure to a size factor, and [Section 4](#) explains why that adjustment could absorb

¹⁰⁴The friction count is heavily right-skewed, with most firms recording little or no friction; see [Figure 8](#).

¹⁰⁵A joint horse-race specification confirms that the two proxies load simultaneously without either absorbing the other, that adding sector and board-structure controls preserves both, and that governance variables carry no independent announcement-day effect; see [Table 16](#) and [Table 23](#).

variation associated with the watch-list; we therefore re-estimate the specification under the market model, which makes no size adjustment. The friction estimate barely moves — a decline of 0.36 percentage points per event, significant at the 1% level — so the penalty does not depend on how the benchmark treats size. The positive watch-list coefficient from the same market-model specification is taken up in [Section 5.3](#).

Table 8: Friction and Announcement Returns (E1)

	FF3					CAPM
	(1)	(2)	(3)	(4)	(5)	(6)
Friction count	-0.0034 ^{***} (0.0013)	-0.0031 ^{**} (0.0012)	-0.0034 ^{**} (0.0016)	-0.0031 ^{**} (0.0012)	-0.0032 ^{**} (0.0016)	-0.0036 ^{***} (0.0011)
Watchlist		0.0524 [*] (0.0316)	0.0575 (0.0360)	0.0512 (0.0327)	0.0559 (0.0370)	0.0323 ^{**} (0.0127)
Above median		0.0227 ^{***} (0.0062)	0.0211 ^{***} (0.0071)	0.0175 ^{***} (0.0067)	0.0203 ^{**} (0.0096)	0.0263 ^{***} (0.0058)
Illiquidity				-0.0018 (0.0098)	-0.0014 (0.0110)	
Catalist				-0.0098 (0.0126)	-0.0001 (0.0176)	
Financial controls	No	No	Yes	No	No	No
Sector FE	No	No	No	No	Yes	No
<i>N</i>	274	274	240	272	243	274
<i>R</i> ²	0.049	0.099	0.135	0.099	0.205	0.140

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the (−1, +3) trading-day window around E1 (February 21, 2025). Columns (1)–(5): Fama–French three-factor model (Asia Pacific ex Japan). Column (6): market model (CAPM) with SGX equal-weighted index, presented jointly because the S\$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

If the penalty reflects lost information, it should be concentrated in the oversight instruments that convey the strongest signals. We therefore split the aggregate friction count by instrument type ([Table 9](#)). The instruments differ in informational content. A query is a public request for clarification, often prompted by unusual trading, whereas a halt or suspension interrupts trading and commonly accompanies regulatory scrutiny or disclosure concerns. Formal regulatory actions are rarer and more severe still. The information account therefore predicts that the penalty should attach primarily to the more serious interventions, while routine queries should carry little independent effect. A penalty distributed evenly across all instruments would instead suggest that the friction count captures a general characteristic of troubled firms rather than the information

conveyed by regulatory oversight.

The estimates support the information account. Halts and suspensions account for the entire statistically reliable penalty: each event predicts a decline of slightly more than one-third of a percentage point, and the estimate changes little as channel and financial controls are added. Query-related friction is statistically indistinguishable from zero in every specification. Regulatory actions carry much larger negative point estimates, ranging from 1.3 to 4.3 percentage points per event, but the large majority of firms record no such action, leaving too little variation for reliable inference. Investors appear to have priced the severity of the regulatory signal rather than the frequency of regulatory contact.

Table 9: Friction Type Decomposition (E1)

	(1)	(2)	(3)
Query-type friction	-0.0025 (0.0023)	-0.0011 (0.0022)	-0.0013 (0.0025)
Halt/suspension-type friction	-0.0036 ^{***} (0.0013)	-0.0038 ^{***} (0.0012)	-0.0036 ^{**} (0.0018)
Regulatory-type friction	-0.0127 (0.0258)	-0.0236 (0.0275)	-0.0433 (0.0381)
Above median		0.0241 ^{***} (0.0064)	0.0227 ^{***} (0.0073)
Watchlist		0.0627 [*] (0.0360)	0.0818 [*] (0.0460)
Financial controls	No	No	Yes
<i>N</i>	274	274	240
<i>R</i> ²	0.056	0.112	0.148

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the (−1, +3) trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Query-type: SGX public queries. Halt/suspension-type: trading halts and suspensions. Regulatory-type: formal regulatory actions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The second decomposition, reported in [Table 10](#), separates friction accumulated in the twelve months before E1 from older friction. This distinction helps clarify how investors used the friction record. If it signalled current compliance problems, the penalty should be concentrated in recent events. If it instead identified firms with a longer history of regulatory concern, the accumulated record should carry greater weight. The estimates support the latter interpretation. Older friction predicts a decline of just under half a percentage point per event and remains marginally significant with and without channel controls. Recent friction is statistically indistinguishable from zero and, if anything, carries a smaller point estimate.

This result creates a puzzle for the adverse-selection account. The firm's earlier queries and suspensions had been public for years, so any information they conveyed should already have been reflected in its share price. An announcement concerning future oversight should not, by itself, cause investors to reprice an old regulatory record. The effect is better understood as forward-looking. The accumulated record identified the firms for which investors had come to rely on continued regulatory monitoring. While the public regime remained in place, even a firm with a troubled history remained observable because renewed problems would generate a fresh query, halt, or suspension. The February announcement proposed to withdraw precisely those signals. That change mattered little for firms with clean histories, but it reduced the expected flow of information about firms whose records suggested a continuing need for scrutiny. Their prices fell accordingly because investors anticipated the loss of future regulatory signals; the historical record itself had already been incorporated into prices.

Table 10: Friction Recency (E1)

	(1)	(2)
Friction count (recent)	-0.0032 (0.0032)	-0.0026 (0.0030)
Friction count (old)	-0.0046* (0.0027)	-0.0044* (0.0025)
Above median		0.0221*** (0.0062)
Watchlist		0.0541* (0.0325)
<i>N</i>	274	274
<i>R</i> ²	0.060	0.108

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the (−1, +3) trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Recent friction: events from March 2024 to January 2025. Old friction: events before March 2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The behaviour of friction at the July manager appointment (E3) supplies a further check. The stacked specification in [Table 11](#) observes each firm at both events and estimates the friction effect separately for E1 and E3. At E1, each additional friction event predicts a decline of approximately 0.3 percentage points. At E3, that effect is fully offset, leaving no relationship between friction and returns. High-friction firms moved with the rest of the market: they neither recovered their February losses nor experienced a further decline, so the E1 markdown remained in place. This pattern is consistent with the information account. E3 concerned the capital programme, including the appointment of managers and disclosure of the mandate terms, and contained no new information about regulatory oversight. There was therefore no new oversight signal

for the market to price. By contrast, the size coefficient reverses at E3 because the mandate disclosure directly concerned capital allocation. Each coefficient thus changes only at the event whose content bears on the corresponding channel. This alignment links the friction penalty specifically to the deregulatory content of E1, rather than to a general tendency for troubled firms to underperform whenever reform is announced.

Two considerations counsel caution about the penalty's size, although neither disturbs its direction. First, the abnormal returns on which the regressions run are themselves estimates from the first-stage benchmark model, and standard errors that treat them as data understate the true uncertainty; under a bootstrap that carries that first-stage estimation error through to the second stage, the friction coefficient retains only marginal significance.¹⁰⁶ Second, most firms carry little accumulated friction, so the estimate rests on comparatively rare events. We therefore place weight on the direction of the penalty, which survives the specification, type, and recency variations (Table 21), rather than on its precise magnitude. Read in this way, the results document the information side of the intermediation gap. The market treated the proposed withdrawal of public oversight as the loss of an informational asset and imposed that loss on the firms previously covered by it.

¹⁰⁶The generated-regressor bootstrap (Section A.2.6) re-estimates the factor loadings and cumulative abnormal returns in each replication, so its standard errors incorporate first-stage estimation noise that conventional HC1 inference does not capture; the E1 friction coefficient remains marginally significant ($p = 0.054$).

Table 11: Stacked E1–E3 Friction Interaction

	(1)	(2)	(3)
Friction count	-0.0034 ^{***} (0.0013)	-0.0029 ^{**} (0.0012)	-0.0033 ^{**} (0.0016)
Above median		0.0214 ^{***} (0.0061)	0.0232 ^{***} (0.0071)
Post-E3	0.0345 ^{***} (0.0065)	0.0572 ^{***} (0.0100)	0.0557 ^{***} (0.0112)
Friction $\times$ Post-E3	0.0040 (0.0026)	0.0031 (0.0025)	0.0044 (0.0032)
Above median $\times$ Post-E3		-0.0387 ^{***} (0.0107)	-0.0397 ^{***} (0.0110)
Financial controls	No	No	Yes
N	544	544	478
R^2	0.118	0.137	0.155

Notes: OLS with event fixed effects and standard errors clustered by stock code, because each firm contributes two observations (one at E1, one at E3); HC1 would understate the standard errors by treating these as independent, when in reality the same firm's CARs at E1 and E3 are correlated. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Post-E3 = 1 for E3 observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Targeted versus Broad Burden Removal

The watch-list is the one systematic exception to the friction penalty, and that exception provides an additional test of the information account. If the market were simply punishing deregulation wherever it touched troubled firms, watch-listed firms, which by definition had recorded losses for at least three consecutive years, should have experienced the largest declines when abolition was proposed at E1. The information account predicts the opposite, precisely because the watch-list differed from the other oversight instruments in what it contributed. As argued in [Section 2](#) and [Section 3.3](#), watch-list entry turned on objective criteria, namely sustained losses and low market capitalisation, that investors could infer from public information. The designation therefore added little new information, while its stigma imposed material costs by impairing access to finance and discouraging counterparties. The restructuring regime in force since 2020 also gave distressed issuers a route to recovery that did not require delisting, increasing the value of removing the designation. Abolishing the watch-list therefore eliminated a substantial burden while sacrificing little information, and its removal should be priced as relief, even while the withdrawal of informative monitoring is priced as a loss.

The E1 estimates are consistent with this prediction. Because watch-listed firms also tend to have

extensive friction records, the comparison must hold overall friction constant; otherwise, the benefit of removing the label would be obscured by the penalty associated with those records. Among firms with comparable levels of overall friction, watch-list status carries a positive coefficient at E1: 5.2 percentage points under the three-factor model, marginally significant, and 3.2 percentage points under the market model, significant at the 5% level. Within a single announcement, the market accordingly priced the two kinds of removal in opposite directions — relief where the instrument imposed stigma while conveying little information, loss where the instrument produced information investors relied on.

The October implementation event (E4) captures the relief when it was realised, rather than merely anticipated. On October 29, 2025, SGX RegCo formally abolished the watch-list and automatically removed all issuers from it. The five previously watch-listed firms in the sample gained 12.7 percent on average under the three-factor model, against 1.0 percent for other firms; under the market model the corresponding figures are 7.3 percent and -1.1 percent (Table 12, Panel A). The regression in Panel B confirms that watch-list status is the only firm-level variable that predicts the E4 response — the pattern one would expect from an event whose content was confined to removing the list, with the gains accruing to the firms the list had burdened and to no one else. The necessary qualification is statistical. Only five firms carried the label at E4, and averages built on five firms are too imprecise for formal tests to separate them from the rest of the sample: the difference in group means is not statistically significant ($p = 0.279$ under the three-factor model, $p = 0.363$ under the market model), and the watch-list coefficient loses significance under inference that accounts for the five-firm treated group ($p = 0.30$ under the three-factor model). We therefore read the E4 magnitudes as indicating direction rather than as precise estimates. That direction completes the contrast that gives this subsection its title: removing a burden narrowly (in this context, a defined label or a known set of firms) produced gains for precisely the firms relieved, whereas the broad withdrawal of monitoring at E1 was priced as a loss across the previously monitored segment.

Table 12: Watch-List Removal Implementation (E4)

<i>Panel A: Mean CARs by watchlist status</i>				
	FF3		CAPM	
	Mean CAR	N	Mean CAR	N
Watchlisted	0.1273 (0.0939)	5	0.0731 (0.0819)	5
Non-watchlisted	0.0098*** (0.0036)	253	-0.0111*** (0.0035)	253
Difference	0.1176 (1.25)		0.0841 (1.03)	
<i>Panel B: Regression</i>				
	FF3		CAPM	
	(1)	(2)	(3)	(4)
Watchlist	0.1176 (0.0844)	0.1110 (0.0847)	0.0841 (0.0737)	0.0817 (0.0740)
Above median		-0.0129* (0.0072)		-0.0048 (0.0071)
N	258	258	258	258
R ²	0.063	0.073	0.036	0.038

Notes: CARs estimated over the $(-1, +3)$ trading-day window around E4 (October 29, 2025). Panel A: Mean CARs under both FF3 (Fama–French three-factor, Asia Pacific ex Japan) and CAPM (market model, SGX equal-weighted index). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for group means; significance is assessed by the Student t -test. Welch t -statistic in parentheses for the difference. Panel B: OLS with HC1 robust standard errors in parentheses. Columns (1)–(2): FF3. Columns (3)–(4): CAPM. CAPM presented jointly because the S\$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. Panel A includes all firms with valid CARs ($N = 263$); Panel B regression sample is $N = 258$ after excluding firms with missing control variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The two sets of results carry different evidential weight, and we assign them different roles in the argument. The friction results, estimated across the full firm sample, provide the principal evidence for the information channel. They show that the market priced the broad withdrawal of oversight as a loss of information, that the penalty is concentrated in severe and long-accumulated forms of scrutiny rather than routine regulatory contact, and that it persists after controlling for firm size, so it cannot be reduced to the capital channel. The watch-list results, based on only five treated firms, serve a narrower corroborative function. They suggest that investors distinguished between a stigmatising classification and informative monitoring, which implies that burden removal could have been more selective: the label could be retired while the instruments supplying useful information were retained. Our conclusions about the information channel do not depend on the watch-list evidence. Taken together, however, the results indicate that investors priced the

informational content of each instrument withdrawn, rather than deregulation in the abstract, consistent with the premise developed in [Section 6](#).

6 Regulatory Design Implications

Our findings frame a familiar policy failure, in which public support does not reach its intended beneficiaries, as a problem of institutional design. The evidence in [Section 5](#) suggests that the reach of a delegated intervention depends less on the scale of the commitment than on the terms governing its implementation: the specificity of the capital mandate in one channel and the selectivity of burden removal in the other.¹⁰⁷ The following subsections examine each design choice and identify the conditions under which the intermediation gap is most likely to bind.

6.1 Designing the Capital Mandate

The February commitment and the July appointment concerned the same total programme size; what changed was the content of the delegation. The February announcement identified neither the managers nor the allocation terms, whereas the July announcement named the managers and required a significant allocation to small- and mid-cap equities. With the committed sum unchanged, the reversal in the size gradient is most naturally attributed to the additional constraints imposed by the mandate. Given the investability constraints documented in [Section 2.1](#), investors initially appear to have assumed that the capital would be deployed in accordance with prevailing institutional preferences. They revised that assessment once the July terms restricted the range of permissible allocations. The central design choice on the capital channel is therefore the specificity of the mandate governing deployment. The commitment signalled the scale of the programme; the mandate determined which firms the market expected it to reach.

Singapore's reform also illustrates how policymakers can create a workable 'allocation expectation'. By this term, we mean a public statement identifying the intended destination of delegated capital. Such an expectation becomes workable when it is sufficiently definite for outside observers to assess a manager's portfolio against it and credible enough for investors to treat it as constraining managerial discretion. Three mutually reinforcing features gave the EQDP allocation expectation credibility. First, the programme identified a target segment of the market, enabling observers to compare each manager's portfolio with the programme's stated purpose. The requirement that appointed managers make a 'significant allocation' to small- and mid-cap equities¹⁰⁸ directed capital towards a defined subset of the SGX-listed universe.

Second, the programme identified the managers responsible for implementing the allocation expectation. Naming Avanda, Fullerton, and J.P. Morgan Asset Management assigned responsibility to identifiable institutions and allowed observers to compare their subsequent portfolios with

¹⁰⁷Dan Awrey and Kathryn Judge, 'Why Financial Regulation Keeps Falling Short' (2020) 61 BC L Rev 2295

¹⁰⁸Monetary Authority of Singapore, 'MAS Appoints First Batch of EQDP Asset Managers' (n 82).

the programme's stated objective. The institutional background of two managers strengthened the signal. Avanda Investment Management was founded by former senior officers of GIC, Singapore's sovereign wealth fund, while Fullerton Fund Management is owned by Temasek, the state investment company. Investors could therefore regard both firms as closely connected to Singapore's public investment institutions and as credible participants in a programme directed towards the domestic equity market.¹⁰⁹

Third, the July disclosure attached the mandate terms to an actual allocation of capital. An allocation expectation that is neither monitored ex post nor backed by consequences for non-compliance remains a weak commitment that investors may discount. The initial S\$1.1 billion placement demonstrated that the mandate was already operating in practice and allowed the market to infer that similar terms would govern the remaining capital under the S\$5 billion programme.

The sequence of the two announcements yields a further design implication. Because credible information about the mandate terms emerged only when the managers were appointed, the market initially priced the programme as leaving managers broad discretion over allocation. The timing of disclosure therefore determines when the constraint enters market prices, making sequencing itself part of programme design. A policy that commits capital well before specifying the terms of deployment creates an interval during which market expectations may diverge from the programme's intended allocation.

The design features that gave Singapore's expectation credibility are, as [Table 13](#) shows, absent from all four comparator programmes. The Bank of Japan's exchange-traded-fund programme, described in [Section 2.1](#), imposed no targeting mandate: purchases tracked broad indices, so the benefits accrued to firms already large enough to qualify for index inclusion, regardless of the programme's scale.¹¹⁰ Korea's Corporate Value-up Programme operated on the issuer side, attaching no allocation mandate to the fund managers who buy issuers' shares, and its reported gains concentrated among larger firms.¹¹¹ China's Government Guidance Funds route public capital through appointed managers, as the EQDP does, but predominantly through venture-capital and private-equity vehicles, where the binding intermediary constraints are those of private-market (i.e., not public-market) deployment.¹¹² The European Union's Listing Act reduces the regulatory costs of accessing public markets without deploying capital, leaving the allocation of capital among listed firms untouched.¹¹³

¹⁰⁹Avanda Investment Management was founded by former GIC senior executives Ng Kok Song, Quah Wee Ghee and Sung Cheng Chih: Avanda Investment Management, 'Overview' (Avanda, 2025); see also 'Ex-GIC CIO's Low-Key Singapore Fund Amasses US\$10b for Global Foray' *The Business Times* (Singapore, 17 February 2022). Fullerton Fund Management Company is wholly owned by Temasek through Sevia Holdings: Temasek, 'Temasek Establishes New Multi-Asset, Multi-Strategy Asset Management Group' (Temasek, 6 October 2020); Fullerton Fund Management, 'About Fullerton' (Fullerton, 2025).

¹¹⁰Charoenwong and others (n 30); Katagiri and others (n 30); Bank of Japan (n 1)

¹¹¹Lee and Peh (n 29); Financial Services Commission (Korea) (n 1)

¹¹²Huang and others (n 26); Ministry of Finance (China) (n 26)

¹¹³the Listing Act (n 2)

Table 13: Comparative Design of Intermediated Equity-Market Programmes

Programme	Channel	Mandate specificity	Target segment	Disclosure timing
Singapore EQDP (2025)	Active mandates to professional fund managers for direct secondary-market purchase of listed equities	Segment-specific: appointed managers expected to make “significant allocation” to small- and mid-cap stocks	Small- and mid-cap SGX-listed equities	Capital committed at announcement (E1, Feb 2025); mandate terms and named managers disclosed later (E3, Jul 2025)
Korea Corporate Value-up (2024)	Voluntary issuer disclosure plus governance incentives and index-linked infrastructure; no direct capital mandate to managers	Issuer-directed, not segment-targeted: firms voluntarily publish value-up plans; no allocation mandate	Listed issuers broadly; the Value-up Index tilts toward larger constituents	Issuer plans disclosed on a voluntary, recurring basis; index launched Sep 2024
Bank of Japan ETF programme (2010–2024)	Passive index replication; purchases executed by an appointed trust bank holding ETFs as trust property	Not a segment-targeting mandate: purchases track broad equity indices for monetary-policy purposes	Index-included (typically larger) firms via TOPIX / Nikkei 225 ETFs	Purchase terms and indices published as monetary-policy operating terms; benefits accrue through index membership
China Government Guidance Funds	Public capital routed through appointed investment managers, primarily venture-capital and private-equity vehicles	Mandate scope varies by fund; oriented to strategic and early-stage sectors rather than listed small-caps	Primary and private-market firms, not secondary-market listed equities	Fund mandates set at establishment; not an announcement-based listed-equity intervention
EU Listing Act (2024)	Supply-side market-structure reform; no public capital deployed through intermediaries	Not an allocation mandate: eases listing and disclosure requirements to lower barriers to public markets	Small and medium-sized enterprises seeking listing	Entered into force Dec 2024; operates through statutory rule change, not periodic mandate disclosure

Notes: The table contrasts intermediated equity-market programmes along the design choices that govern whether public support reaches an intended segment. “Channel” records how the programme transmits support; “Mandate specificity” records whether the instrument attaches a segment-specific allocation expectation to the deploying intermediary; “Target segment” records the firms the programme is configured to reach; and “Disclosure timing” records when the operative terms become public relative to commitment. The Singapore EQDP is distinctive in pairing an active, direct-purchase mandate with a segment-specific allocation expectation, with the operative terms disclosed at E3 after capital was committed at E1. The Bank of Japan programme transmits support through passive index replication, so its incidence is determined by index membership rather than by a targeting mandate. Korea’s Value-up programme works through voluntary issuer disclosure and index-linked infrastructure rather than a capital mandate to managers. China’s Government Guidance Funds and the EU Listing Act are included for contrast: the former operates through private-market vehicles rather than secondary-market purchases of listed equities, and the latter is a supply-side rule reform that deploys no public capital. The public mandate formulation is “significant allocation” to small- and mid-cap equities (MAS media release, 21 July 2025). Programme details for the Bank of Japan and Korea facilities draw on the sources cited in the text: Charoenwong and others and Katagiri and others for the Bank of Japan, and Lee and Peh for Korea, together with the official programme announcements (n 1).

Mandate specificity nonetheless has a cost, because the discretion it constrains serves an important screening function. Professional managers may avoid thin, illiquid, and informationally opaque firms because those characteristics reflect genuine differences in risk, and identifying such risks is part of the purpose of delegating investment decisions to them. A mandate that constrains

this discretion too tightly, whether by prescribing allocation shares in detail or by limiting the manager's ability to exclude the weakest firms, may reintroduce the risks that delegation is intended to manage.¹¹⁴ The design problem is therefore to create a credible expectation of exposure to the target segment while preserving sufficient discretion over the selection of individual securities. The extent to which any particular mandate closes the intermediation gap remains an empirical question. The approximately 0.7 percentage point advantage retained by larger firms across the two events (Section 5.1) is consistent with mandate terms that redirected capital towards the target segment while leaving managers latitude over which firms within that segment to hold. On this interpretation, the partial narrowing of the gap suggests that the programme moved towards such a balance without eliminating intermediary discretion.

6.2 Information-Preserving Deregulation

On the information channel, the design question is which oversight instruments to remove. As Section 2 distinguished, the instruments the reform retired were of two kinds. The public queries, halts, and suspensions produced information: each added to the public record that a particular firm had drawn regulatory scrutiny, and investors used that record to price firms whose quality they could not easily verify. The watch-list label worked differently. Its entry criteria were computable from published information — a firm joined the list when sustained losses coincided with low market capitalisation — so the label added little to what the market already knew, while its stigma imposed real financing and counterparty costs on the firms that carried it. Section 5 showed that the market priced the two kinds of removal in opposite directions: firms with heavy records of queries, halts, and suspensions were marked down when the February announcement proposed to withdraw the oversight apparatus, whereas watch-listed firms, once their overall friction is held constant, gained at that same announcement, and the previously watch-listed firms gained sharply relative to others when the label was abolished in October. Wholesale deregulation therefore combines two effects: relief from regulatory burdens and the loss of information supplied by oversight. A more selective approach could remove instruments that primarily impose burdens while preserving those that continue to produce information valued by investors. We refer to this design choice as the selectivity of burden removal.

SGX RegCo's replacement of public queries with private engagement provides the clearest example of an informative instrument being withdrawn. Under the former regime, a query was published when the exchange directed it to an issuer, usually after detecting unusual trading, so all investors learned that the regulator had identified a firm-specific concern. Under the new regime, the exchange may continue to make the same enquiry, but does so privately. Supervision therefore remains in place, while the market loses visibility into which firms are receiving regulatory attention. Our event evidence does not identify a contemporaneous cost from the removal of public queries, and the reason is important. The friction regressions measure how investors priced each firm's accumulated history of regulatory scrutiny, within which past queries carried no

¹¹⁴He and Xiong (n 17). The trade-off between issuing an outright command and leaving real authority to a better-informed agent is analysed by Philippe Aghion and Jean Tirole, 'Formal and Real Authority in Organizations' (1997) 105 J Pol Econ 1.

detectable penalty. However, the relevant loss is prospective: queries that would previously have entered the public record will instead occur, if at all, through private engagements that investors cannot observe. The reform therefore removes a future stream of information, not a cost already reflected in the firm's historical friction record. A regulator that withdraws a public oversight instrument on this basis must accordingly consider what mechanism, if any, will replace the information it previously supplied.

The substitutes available to a regulator in this situation differ in how much of the query's content they preserve. The regulator could publish periodic reports identifying the firms with which it has engaged privately, thereby revealing where scrutiny has fallen without flagging each concern in real time. It could disclose each engagement after the matter has been resolved, preserving the completeness of the public record at the cost of timeliness. Alternatively, it could require the issuer to announce that it has received a regulatory enquiry, preserving the contemporaneous signal while shifting the disclosure obligation from the regulator to the firm.¹¹⁵ Each approach would restore part of what the public query supplied, namely notice that a particular firm had attracted regulatory attention, while avoiding a contemporaneous public warning issued by the regulator itself.

The third option exposes a premise on which the whole disclosure-based regime rests — that issuer disclosure can assume the informational role previously performed by public oversight. This premise is demanding. Disclosure informs investors only when they notice it and understand it, and the experimental evidence on disclosure efficacy shows that these conditions often fail, especially for unsophisticated recipients who do not fully incorporate disclosed information into their decisions.¹¹⁶ These limitations are especially important in a market with a substantial retail-investor base, as Singapore has. Withdrawing public oversight shifts the burden of assessing firm quality onto investors. A retail investor must locate and assess the issuer's own disclosures, whereas a public query provides a salient and verifiable signal that the regulator has identified grounds for scrutiny. That signal is especially valuable where investor compensation and private avenues of recourse remain limited, as they do in Singapore despite ongoing reform efforts.¹¹⁷ Information-preserving deregulation therefore imposes two demanding tasks on the regulator: identifying the oversight instruments on which investors rely when pricing firms, and providing an adequate substitute when any such instrument is withdrawn.

¹¹⁵The closest natural experiment arises when a regulator moves a previously confidential oversight channel into public view: Amy P Hutton, Susan Z Shu and Xin Zheng, 'Regulatory transparency and the alignment of private and public enforcement: Evidence from the public disclosure of SEC comment letters' (2022) 145 J Fin Econ 297, finding that the disclosure realigned private and public enforcement and altered the affected firm's information environment.

¹¹⁶Geneviève Helleringer, 'Designing Disclosures: Testing the Efficacy of Disclosure in Retail Investment Advice' in Klaus Mathis and Avishalom Tor (eds), *Nudging — Possibilities, Limitations and Applications in European Law and Economics* (Springer 2016) 153

¹¹⁷In other words, when investors have little prospect of recovering losses ex post through litigation, ex ante signals that a firm has drawn regulatory attention becomes correspondingly more valuable. Securities class actions are not permitted in Singapore, and claims for investor compensation for misstatement remain rare: Alexander FH Loke, 'The Efficacy of Securities Investors' Rights in Singapore' [2009] Sing JLS 109; Wan, Chen and Goo (n 74). However, MAS is now consulting on measures to facilitate collective action and civil compensation for market misconduct: see the MAS investor-recourse consultation (n 75).

6.3 Boundary Conditions and the Integrated Regulator

The intermediation gap documented here depends on institutional structure and is therefore best understood through a set of boundary conditions. Three are especially important — two concern the capital channel, while the third concerns the information channel. The gap should be strongest where all three are present. First, the intended beneficiaries must fall below the size threshold at which professional managers are willing to hold a firm’s stock. At that threshold, the allocation decision becomes effectively binary: either the issuer enters the investable universe or the delegated capital does not reach it. Capital therefore stops at the threshold instead of declining gradually across firms. Singapore’s size gradient displays exactly this threshold pattern, as [Section 5.1](#) documents.

Second, capital must be deployed through a small group of managers subject to similar portfolio constraints. Where managers are numerous and their mandates heterogeneous, their allocation choices may offset one another. Where they are few and face similar constraints, those choices are more likely to move together, turning a common preference for larger firms into a market-wide demand effect. Singapore’s initial placement involved only three appointed managers, whose similar constraints could shift capital in the same direction.

Third, the relevant supervision must be visible to the market. The information channel operates through signals that investors can observe, so withdrawing an oversight instrument causes informational loss only where that instrument was public. In Singapore, queries, trading halts, and suspensions are individually disclosed, allowing investors to incorporate the resulting record of regulatory scrutiny into prices. The proposed withdrawal of those instruments was therefore capable of being priced as a loss of information. Where supervision is conducted privately, there is little public signal to remove, and an otherwise similar deregulatory change should have a correspondingly smaller effect on market prices.

The same features that make Singapore a useful setting for identifying the intermediation gap also limit the extent to which the findings can be generalised. In deeper markets, the two conditions governing the capital channel may weaken in tandem. Analyst coverage and institutional ownership extend further down the size distribution, so the intended beneficiaries of a support programme are more likely to fall within the investable universe; the sharp threshold documented in [Section 5.1](#), at which delegated capital may cease to reach smaller firms, is therefore less pronounced. Capital is also distributed across a larger and more heterogeneous group of managers, making it less likely that a common portfolio constraint will redirect aggregate demand towards the largest firms. The gap should accordingly be strongest in markets that share Singapore’s institutional features, including smaller European growth boards, Hong Kong’s Growth Enterprise Market (GEM), and mid-tier Asian exchanges, and should weaken as market depth increases and intermediation becomes more dispersed. The magnitude of the effect in other settings, and whether greater mandate specificity would redirect capital in deeper markets where these conditions hold only weakly, remain questions for future research on comparable programmes.

A final design question concerns the institutional architecture through which the reform was developed. In Singapore, the two problems examined in this section — specifying the capital

mandate and deciding which regulatory burdens to remove — were addressed within a single integrated framework. MAS introduced the developmental instrument, while SGX RegCo, acting as its supervisory delegate, introduced the deregulatory measures, and both formed part of the same reform package.¹¹⁸ This architecture matters because it determines where the trade-off between development and supervision is assessed. An integrated regulator can weigh the gains from lighter oversight against the informational costs imposed on the market it regulates. Where functions are divided among specialised agencies, as under a ‘twin-peaks structure’ that assigns prudential regulation and market-conduct regulation to separate bodies, the trade-off must instead be coordinated across institutional boundaries. Indeed, our announcement-return evidence shows how the market priced the instruments produced by Singapore’s integrated framework. It does not reveal how MAS and SGX RegCo deliberated internally or how those deliberations shaped the final design. Our analysis therefore cannot determine whether integration resolves the two design problems more effectively than institutional separation. That question requires comparative evidence from jurisdictions in which the relevant functions are divided.¹¹⁹

7 Conclusion

In this article, we have examined Singapore’s 2025 equity-market reform, which was implemented through two arms of an integrated regulatory architecture. MAS introduced the capital instrument, while SGX RegCo recalibrated the supervisory regime. We have treated the central problem as one of transmission: whether each instrument reached the firms it was intended to benefit. Although the capital commitment targeted small- and mid-cap equities, investors initially priced it as a large-cap intervention. That relationship reversed, and the initial gap narrowed, only after the disclosed mandate terms required significant allocation to the target segment. On the supervisory side, we find that investors distinguished between the instruments deregulation removed. The market priced the broad withdrawal of informative oversight as a loss concentrated among firms with extensive regulatory histories, while treating the removal of a stigmatising classification as relief. We use the term ‘regulatory intermediation gap’ to describe the design problem that arises when public intervention is transmitted through private managers and regulatory institutions on terms that prevent it from reaching its intended beneficiaries. Across the sequence of announcements, prices changed when the content of those terms and signals changed. The gap therefore depends in part on design choices that remain within the regulator’s control.

Our argument extends beyond Singapore to interventions that channel public capital through private intermediaries while reducing public oversight. We expect the gap to bind most strongly where the intermediaries deploying corrective capital face the same constraints that produced the original capital deficit, and where deregulation withdraws public signals on which investors rely to price opaque firms. These conditions are especially plausible in markets dominated by a

¹¹⁸For an examination of the tension between a single agency’s developmental or competitiveness mandate and its supervisory mandate, see Eilís Ferran, ‘International Competitiveness and Financial Regulators’ Mandates: Coming Around Again in the UK’ (2023) 9 J Fin Reg 30.

¹¹⁹John Armour and others, *Principles of Financial Regulation* (Oxford University Press 2016)

small number of large issuers, where asset management is concentrated and regulatory scrutiny is publicly visible. We would expect the gap to weaken as intended beneficiaries move inside the investable universe, intermediation becomes more dispersed, and supervision generates fewer public signals.

Our analysis identifies two design choices that warrant attention alongside the scale of the intervention: the specificity of the mandate attached to delegated capital and the selectivity with which regulatory burdens are removed. A mandate must constrain intermediaries sufficiently to redirect capital towards the intended segment while preserving their ability to screen individual firms. Similarly, deregulation must distinguish instruments that principally impose costs from those that continue to supply information valued by investors. These choices help determine whether public intervention reaches its intended beneficiaries and what is lost in the process. Because the evidence comes from a relatively small and illiquid equity market, further research should test whether the regulatory intermediation gap persists in deeper markets, where capital is deployed through more heterogeneous channels and regulatory signals are less publicly salient.

A Robustness and Supplementary Results

This appendix gathers the identification tests, the robustness checks, and the supplementary event-level results that the main text relies on but does not display.

A.1 Identification and Placebo Tests

A.1.1 Paired Reversal Placebo

As discussed in [Section 5.1](#), the reversal in the size coefficient between E1 and E3 supplies much of the evidence for the intermediation gap hypothesis. The first question is whether a reversal of that magnitude could arise by chance. To address that question, we construct a paired-date reversal placebo. We draw two hundred pairs of pseudo-event dates uniformly from June 2023 to January 2025, requiring each pair to be separated by about 105 trading days, the span corresponding to the E1–E3 gap, and excluding the window around E0. For each pair, we estimate firm-level CARs at both dates over a standard $[-250, -30]$ estimation window and compute the size and friction coefficients from the channel-separation horse race in [Table 16](#). The resulting reversal statistic, $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$, records how much each coefficient changes across the two pseudo-events.

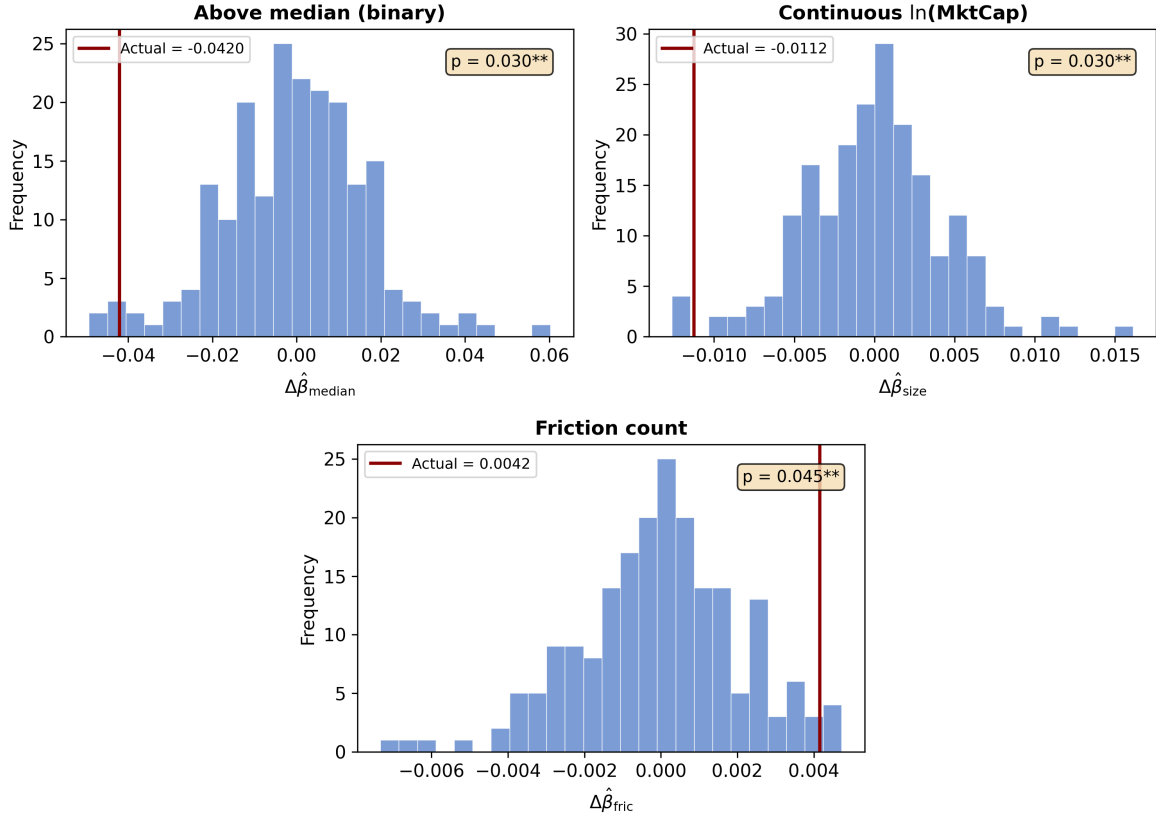
As shown in [Table 14](#), the observed binary above-median reversal lies at the 2nd percentile of the placebo distribution, with $\Delta\hat{\beta}_{\text{median}} = -0.042$ ($p = 0.030$); only 6 of the 200 placebo pairs generate a reversal as large. The continuous size measure reaches the same tail of the distribution ($\Delta\hat{\beta}_{\text{size}} = -0.011$, 2nd percentile, $p = 0.030$). At the other end of the distribution, the friction coefficient reversal ($\Delta\hat{\beta}_{\text{fric}} = 0.004$, 98th percentile, $p = 0.045$) is likewise too extreme to attribute to chance. [Figure 6](#) plots the three placebo distributions and marks the observed reversals.

Table 14: Paired Reversal Placebo (200 Pseudo-Event Pairs)

Variable	Actual $\Delta\hat{\beta}$	Placebo Mean	Placebo SD	Pctile	Perm. p
Above median	−0.0420	−0.0007	0.0169	2.0%	0.030 ^{**}
log(Market cap)	−0.0112	−0.0001	0.0044	2.0%	0.030 ^{**}
Friction count	+0.0042	−0.0000	0.0021	98.0%	0.045 ^{**}

Notes: 200 pseudo-event date-pairs separated by ≈ 105 trading days (matching the E1–E3 gap). $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ from the channel-separation horse race ([Table 16](#)) with HC1 robust standard errors. Two-sided p : fraction of placebos with $|\Delta\hat{\beta}| \geq |\Delta\hat{\beta}_{\text{actual}}|$. Above median = 1 if $\ln(\text{MktCap})_i \geq \text{sample median}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Distribution of placebo coefficient reversals from 200 random date-pairs. Vertical lines mark the observed E1-to-E3 reversals.



Note: Top-left panel: distribution of $\Delta\hat{\beta}_{\text{median}}$ (binary above-median coefficient) across 200 placebo pairs; the actual reversal (-0.0420 , maroon line) falls at the 2nd percentile ($p = 0.030$). Top-right panel: distribution of $\Delta\hat{\beta}_{\text{size}}$ (continuous $\ln(\text{Market cap})$ coefficient); the actual reversal (-0.0112) falls at the 2nd percentile ($p = 0.030$). Bottom panel: distribution of $\Delta\hat{\beta}_{\text{fric}}$ (friction count); the actual reversal (0.0042) falls at the 98th percentile ($p = 0.045$). All three reversals are statistically significant at the 5% level.

A.1.2 Cross-Market Placebo: Hong Kong

The Hong Kong placebo provides a neighbouring-market check on the size gradient documented for SGX firms: if that gradient merely reflects a regional size effect, matched HKEX firms should display a comparable pattern despite the absence of any reform. To construct Figure 4, we match each SGX firm to its three nearest HKEX neighbours by log market capitalisation ($K = 3$), yielding a matched sample of 644 Hong Kong firms, and then estimate the same size-CAR specification on both markets at E1. Panel A of Table 15 reports the continuous $\ln(\text{MktCap})_i$ specification alongside the binary above-median indicator. Under the continuous measure, the SGX coefficient is positive and significant at the 1% level (0.006), whereas the matched HK coefficient is negative and marginally significant (-0.004); the cross-market difference is itself significant ($p = 0.001$). The binary indicator sharpens the same contrast to the 1% level. Panel B then turns from the regression specification to size-quartile CARs within each market. Within SGX, the quartile CARs

rise steadily from $Q1$ to $Q4$, while the HK sample runs in the opposite direction, producing a significantly negative size gradient and a $Q4-Q1$ spread of -2.78 percentage points ($p = 0.024$).

Table 15: Cross-Market Placebo: SGX vs Matched Hong Kong Sample (E1)

	SGX	HK (matched)	Difference	p
<i>Panel A: Size regressions (E1, FF3 CAR[−1, +3])</i>				
log(Market cap)	+0.0057 ^{***} (0.0020)	−0.0040 [*] (0.0021)	+0.0097 ^{***} (0.0029)	0.001
Above median	+0.0248 ^{***} (0.0071)	−0.0166 [*] (0.0088)	+0.0414 ^{***} (0.0113)	0.000
N	272	644	916	
R^2	0.035	0.005		
<i>Panel B: Size quartile CARs (% , E1)</i>				
	SGX		HK (matched)	
	Mean CAR	N	Mean CAR	N
Q1 (smallest)	−2.75 ^{***}	68	−0.41	161
Q2	−2.33 ^{***}	68	−1.96 ^{**}	161
Q3	−0.34	68	−2.51 ^{***}	161
Q4 (largest)	+0.22	68	−3.19 ^{***}	161
Q4 − Q1	+2.98 ^{***} [$t = 2.74$] ($p = 0.007$)		−2.78 ^{**} [$t = -2.26$] ($p = 0.024$)	
Cross-market Q4−Q1 difference: +5.75 pp ^{***} ($p = 0.000$)				

Note: Panel A reports OLS regressions of FF3 CARs (−1, +3) on size measures. The HK sample consists of 644 HKEX firms matched to SGX firms by log market capitalisation ($K = 3$ nearest neighbours). HC1 robust standard errors in parentheses. The difference column tests $H_0: \beta_{SGX} = \beta_{HK}$ using $z = (\hat{\beta}_{SGX} - \hat{\beta}_{HK}) / \sqrt{SE_{SGX}^2 + SE_{HK}^2}$. Panel B reports mean CARs by size quartile within each market. ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

A.2 Specification Robustness

A.2.1 Channel-Separation Horse Race

The horse race tests whether the two channel proxies survive joint estimation. Table 16 reports the estimates, with controls added in stages. Entered jointly, the above-median size indicator (0.018) and the friction count (−0.0031) each remain significant, and neither absorbs the other; adding sector fixed effects and board-structure controls leaves both in place. Board independence

and CEO-chair separation carry no independent announcement-day effect, which is consistent with the univariate governance regressions across events (Table 23). The size–friction correlation, at only $\rho = -0.21$ (Figure 7), is low enough for the joint regression to distinguish the two effects.

Table 16: Channel Separation: Joint Specification (E1)

	(1)	(2)	(3)	(4)	(5)
Above median	0.0249 ^{***} (0.0065)		0.0175 ^{***} (0.0067)	0.0203 ^{**} (0.0096)	0.0193 ^{***} (0.0072)
Friction count		-0.0034 ^{***} (0.0013)	-0.0031 ^{**} (0.0012)	-0.0032 ^{**} (0.0016)	-0.0030 ^{**} (0.0013)
Watchlist			0.0512 (0.0327)	0.0559 (0.0370)	0.0508 (0.0329)
Illiquidity (z-Amihud)			-0.0018 (0.0098)	-0.0014 (0.0110)	-0.0012 (0.0099)
Catalist			-0.0098 (0.0126)	-0.0001 (0.0176)	-0.0055 (0.0129)
Independence share					-0.0195 (0.0244)
No CEO duality					0.0085 (0.0102)
Sector FE	No	No	No	Yes	No
N	274	274	272	243	260
R^2	0.051	0.049	0.099	0.205	0.096

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq \text{sample median}$. Spec (4) includes SSIC level-1 sector fixed effects. Spec (5) adds board independence share and CEO-chair separation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.2 CAR Window Robustness

Table 17 re-estimates the E1 horse race specification across five CAR windows, asking whether the estimates depend on the timing convention. The above-median size indicator remains significant at the 5% level in all five specifications, ranging from 0.011 at $(-1, +1)$ to 0.025 at $(-1, +5)$. Whether the size effect operates as a threshold or as a smooth gradient is a separate question, addressed with the continuous size measure in Section A.2.5. The friction coefficient is similarly stable across windows: it equals -0.0031 for the baseline $(-1, +3)$, -0.0034 at $(-1, +5)$, and -0.0027 at $(0, +2)$,

with each estimate significant at the 5% level. The narrowest window $(-1, +1)$ yields a smaller friction coefficient (-0.0017) , which remains significant at the 5% level. This pattern suggests that friction-related information was incorporated into prices over several trading days rather than on the announcement day alone.

Table 17: CAR Window Robustness: E1 Horse Race

	$(-1, +1)$	$(-1, +3)$	$(-1, +5)$	$(0, +2)$	$(0, +5)$
Above median	0.0112** (0.0052)	0.0175*** (0.0067)	0.0250*** (0.0089)	0.0112** (0.0051)	0.0214** (0.0083)
Friction count	-0.0017** (0.0008)	-0.0031** (0.0012)	-0.0034** (0.0016)	-0.0027** (0.0011)	-0.0024* (0.0015)
Watchlist	0.0282 (0.0192)	0.0512 (0.0327)	0.1111** (0.0485)	0.0610* (0.0325)	0.0914** (0.0444)
Illiquidity	0.0023 (0.0038)	-0.0018 (0.0098)	-0.0076 (0.0112)	-0.0088 (0.0100)	-0.0088 (0.0118)
Catalist	-0.0182* (0.0100)	-0.0098 (0.0126)	-0.0063 (0.0141)	-0.0028 (0.0094)	-0.0012 (0.0124)
N	267	272	274	269	274
R^2	0.099	0.099	0.101	0.113	0.076

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the indicated trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq \text{sample median}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.3 Factor Model Robustness

Table 18 reports the E1 horse race specification under three return benchmarks: CAPM, FF3, and the Fama–French five-factor model (FF5).¹²⁰ Across these specifications, the above-median coefficient is significant at the 1% level under CAPM (0.020) and FF3 (0.018), and at the 5% level under FF5 (0.020). The friction coefficient also remains negative and stable, taking values of -0.0036 under CAPM, where it is significant at the 1% level, and -0.0031 under FF3 and -0.0029 under FF5, each significant at the 5% level. The CAPM friction coefficients run about 16% larger than their FF3 counterparts because the size–friction overlap lets the Asia Pacific ex Japan SMB factor absorb part of the friction effect. Although the watch-list coefficient is significant at the 5% level under CAPM (0.030), it weakens to insignificance under FF3 (0.051) and FF5 (-0.012), again because SMB absorbs the relevant variation.

¹²⁰Eugene F Fama and Kenneth R French, ‘A Five-Factor Asset Pricing Model’ (2015) 116 J Fin Econ 1

Table 18: Factor Model Robustness: E1 Horse Race

	CAPM	FF3	FF5
Above median	0.0204 ^{***} (0.0064)	0.0175 ^{***} (0.0067)	0.0198 ^{**} (0.0086)
Friction count	-0.0036 ^{***} (0.0011)	-0.0031 ^{**} (0.0012)	-0.0029 ^{**} (0.0013)
Watchlist	0.0297 ^{**} (0.0137)	0.0512 (0.0327)	-0.0122 (0.0318)
Illiquidity	-0.0015 (0.0030)	-0.0018 (0.0098)	0.0281 ^{***} (0.0059)
Catalist	-0.0124 (0.0115)	-0.0098 (0.0126)	-0.0081 (0.0162)
N	272	272	234
R^2	0.143	0.099	0.116

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1. CAPM: market model with SGX equal-weighted index. FF3: Fama–French three-factor model (Asia Pacific ex Japan, primary specification). FF5: Fama–French five-factor model (Asia Pacific ex Japan). The CAPM friction coefficient is $\approx 16\%$ larger than FF3 because the APXJ SMB factor absorbs part of the friction–size overlap. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.4 Size Effect Across Alternative Measures

Panel A of Table 19 reports the univariate size regressions by event: the E1 coefficient is significant at the 1% level (0.0054), while the E3 coefficient is significant at the 5% level (-0.0059). Panel B then carries the reversal across four alternative size measures — log market cap, log assets, the above-median indicator, and the composite z -score — each of which shows the same E1-positive, E3-negative pattern.

Table 19: Size Effect Across Alternative Measures

	E1	E2	E3		
<i>Panel A: Univariate</i>					
log(Market cap)	0.0054*** (0.0018)	-0.0012 (0.0018)	-0.0059** (0.0026)		
<i>N</i>	272	273	269		
<i>R</i> ²	0.038	0.002	0.024		
<i>Panel B: Alternative measures (E1 vs. E3 sign flip)</i>					
	E1 β	E1 <i>p</i>	E3 β	E3 <i>p</i>	Sign Flip
log(Market cap)	+0.00540	(0.003)	−0.00585	(0.024)	Yes
log(Assets)	+0.00309	(0.074)	−0.00542	(0.036)	Yes
Above median	+0.02492	(0.000)	−0.01751	(0.074)	Yes
Top decile					
z-size (reversed)	−0.00980	(0.018)	+0.01523	(0.013)	Yes

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Panel B: Univariate regressions per measure. z.size sign reversed (higher = smaller). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.5 Continuous Size Measure

Table 20 re-estimates the horse race specification with the continuous $\ln(\text{MktCap})_i$ measure in place of the binary above-median indicator. The size coefficient is positive throughout, though with wider confidence intervals than the binary measure: 0.0032 in the joint specification, where it is marginally significant, and 0.0035 in the governance-augmented specification, where it reaches significance at the 5% level. These wider standard errors trace to the concavity of the size–CAR relationship documented in Figure 3, where the relationship is steep below the median and flat above it, which favours a threshold specification over a log-linear one. All remaining coefficients are virtually unchanged between the continuous and binary specifications.

Table 20: Horse Race with Continuous Size Measure (E1)

	(1)	(2)	(3)	(4)	(5)
log(Market cap)	0.0054*** (0.0018)		0.0032* (0.0017)	0.0036 (0.0023)	0.0035** (0.0017)
Friction count		-0.0034*** (0.0013)	-0.0032*** (0.0012)	-0.0032** (0.0015)	-0.0031** (0.0013)
Watchlist			0.0505 (0.0327)	0.0560 (0.0390)	0.0498 (0.0330)
Illiquidity (z-Amihud)			-0.0017 (0.0095)	-0.0014 (0.0107)	-0.0010 (0.0095)
Catalist			-0.0114 (0.0124)	-0.0018 (0.0179)	-0.0074 (0.0127)
Independence share					-0.0184 (0.0249)
No CEO duality					0.0091 (0.0101)
Sector FE	No	No	No	Yes	No
<i>N</i>	272	274	271	243	260
<i>R</i> ²	0.038	0.049	0.087	0.195	0.083

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Same specification as Table 16 but using continuous $\ln(\text{MktCap})_i$ instead of the binary above-median indicator. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.6 Generated-Regressor Bootstrap

Because the dependent variable is itself an estimate, the second-stage standard errors may understate true uncertainty.¹²¹ To address that concern, we run a two-stage bootstrap of $B = 2,000$ replications over a pre-E1 estimation window of $(-250, -30)$. The first stage applies a moving-block bootstrap (block length 5 trading days) to each firm’s joint factor-and-return time series, re-estimating Maynes–Rumsey factor loadings and recomputing CARs at the original event window. The second stage applies a pairs bootstrap over firms for the E1 horse race and a cluster-pairs bootstrap, with clusters defined by stock code, for the stacked E1–E3 specification. The bootstrap distribution thereby absorbs both the first-stage estimation noise and the residual variation across firms.

Table 21 reports bootstrap standard errors and 95% percentile intervals, placing them against the

¹²¹Pagan (n 98)

conventional HC1 (E1 horse race) and CR1 (stacked reversal) baselines. Although the standard errors widen under the bootstrap, the two headline coefficients remain significant at the 1% level: the E1 above-median size effect ($p = 0.007$, SE wider by 27%) and the E1-to-E3 reversal γ_2 ($p = 0.001$, SE wider by 17%). The friction coefficient is more fragile: it weakens to marginal significance ($p = 0.054$), exposing first-stage noise that HC1 inference does not register. Finally, the FF3 watch-list coefficient is not statistically significant under bootstrap inference ($p = 0.30$), which is consistent with the five-firm treatment cell and the SMB-absorption issue already discussed in the factor-robustness appendix.

Table 21: Generated-Regressor Bootstrap ($B = 2,000$)

	Coef.	Conv. SE	Conv. p	Boot SE	Boot 95% CI	Boot p
AboveMedian _{i} at E1 (Table 8)	0.0227***	(0.0062)	0.000	0.0079***	[+0.0087, +0.0404]	0.007
FricCount _{i} at E1 (Table 8)	-0.0031**	(0.0012)	0.011	0.0016*	[-0.0065, -0.0002]	0.054
Watchlist _{i} at E1 (Table 8)	0.0524*	(0.0316)	0.097	0.0543	[-0.0536, +0.1644]	0.304
AboveMedian _{i} $\times \mathbb{1}_{e=E3}$ reversal (Table 6)	-0.0424***	(0.0116)	0.000	0.0136***	[-0.0643, -0.0108]	0.001

Notes: Two-stage bootstrap with $B = 2,000$ replications. First stage: moving-block bootstrap (block length 5 trading days) of each firm's joint factor-and-return time series within the pre-E1 estimation window (-250, -30); Maynes-Rumsey factor loadings re-estimated and CARs recomputed at the original (-1, +3) event window. Second stage: pairs bootstrap over firms (E1 horse race) or cluster-pairs bootstrap on stock code (stacked E1-E3 specification). "Conv. SE" is HC1 (E1 horse race) or CR1 clustered on stock code (stacked reversal). "Conv. p " is the two-sided asymptotic p -value implied by the conventional standard error; stars on the coefficient column follow this p -value. "Boot 95% CI" is the 2.5%-97.5% percentile interval of the bootstrap distribution. "Boot p " is the two-sided percentile p -value against H_0 : coefficient = 0; stars on the bootstrap SE column follow this p -value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.7 Catalyst Board-Tier Effect

The Catalyst board-tier coefficient reported in preliminary analyses was significant at the 1% level (-3.0 percentage points), but it turns out to be an artefact of sample composition. The initial sample included non-equity stock codes (warrants, structured products, daily leverage certificates), all classified as Mainboard. Their positive mean CARs at E1 inflated the Mainboard reference group and exaggerated the apparent Catalyst underperformance. Table 22 reports the equity-only results. Under CAPM, the bare Catalyst coefficient is -2.82 percentage points, significant at the 1% level (Panel A). The FF3 model, which adds the SMB factor, reduces it to -2.44 percentage points, significant at the 5% level, and adding the above-median size indicator carries it further, to -1.23 percentage points and insignificance. Because Catalyst firms are on average smaller than Mainboard firms, the board-tier variable mainly proxies for size.

Table 22: Catalyst Board-Tier Effect

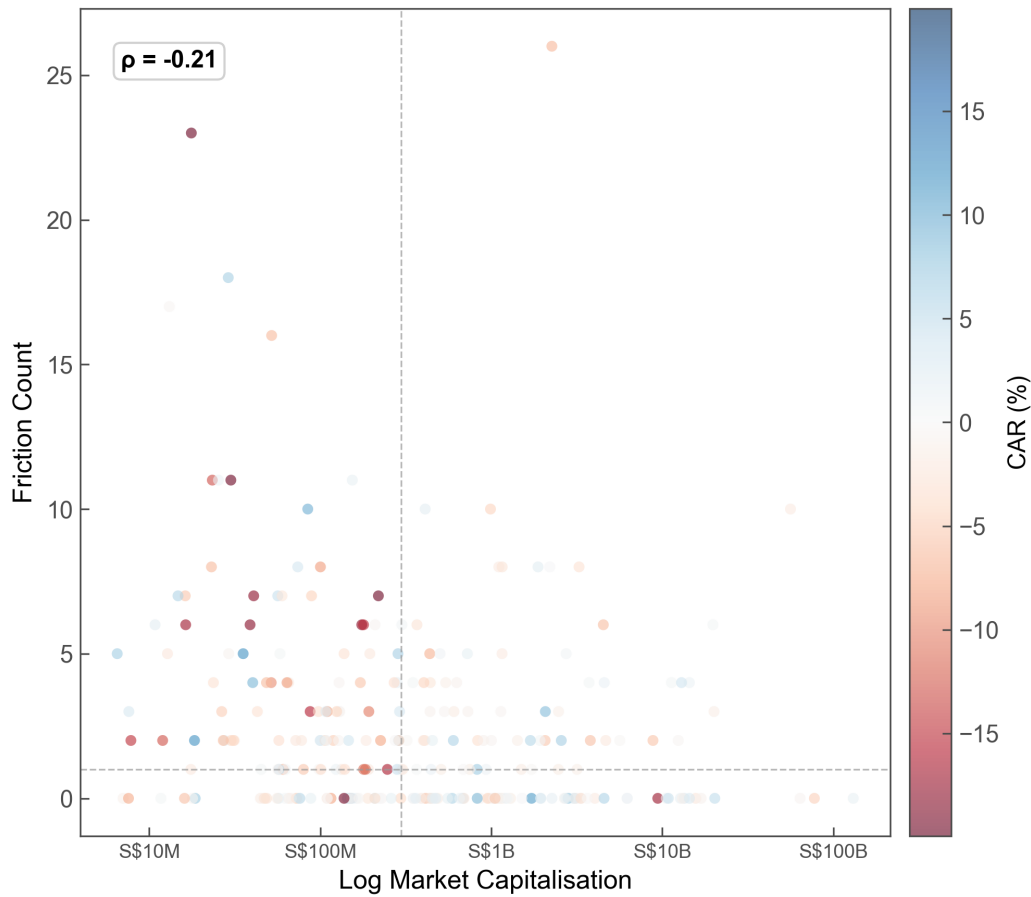
<i>Panel A: Catalyst coefficient by model and controls (E1)</i>				
	FF3		Market Model	
	Bare	+Size	Bare	+Size
Catalist	-0.0244** (0.0121)	-0.0123 (0.0132)	-0.0282*** (0.0106)	-0.0137 (0.0117)
Above median		0.0214*** (0.0071)		0.0257*** (0.0068)
<i>N</i>	274	274	274	274
<i>R</i> ²	0.025	0.057	0.037	0.087
<i>Panel B: Board means by event (FF3 CAR(−1, +3) %)</i>				
	E1	E2	E3	
Mainboard	-0.95 (0.33)	-0.23 (0.28)	+2.65 (0.44)	
Catalist	-3.39 (1.18)	+0.55 (1.30)	+5.61 (1.66)	
Diff (Cat−MB)	-2.44* (-2.00)	+0.77 (0.58)	+2.96* (1.72)	

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the (−1, +3) trading-day window around E1. FF3: Fama–French three-factor model (Asia Pacific ex Japan). Market Model: CAPM with SGX equal-weighted index. The FF3 size factor (SMB) absorbs most of the board effect. Panel B: Mean CARs (%) by board tier (equity-only sample). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for board means; significance is assessed by the Student *t*-test. Welch *t*-statistic in parentheses for the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.8 Size–Friction Correlation

Figure 7 provides the visual analogue to the horse-race regressions in Table 16: it plots log market capitalisation against the regulatory friction count, the two variables entered jointly in those regressions.

Figure 7: Scatter plot of log market capitalisation against regulatory friction count. The negative correlation (Pearson $\rho = -0.21$) confirms that size and friction are empirically related but not collinear.

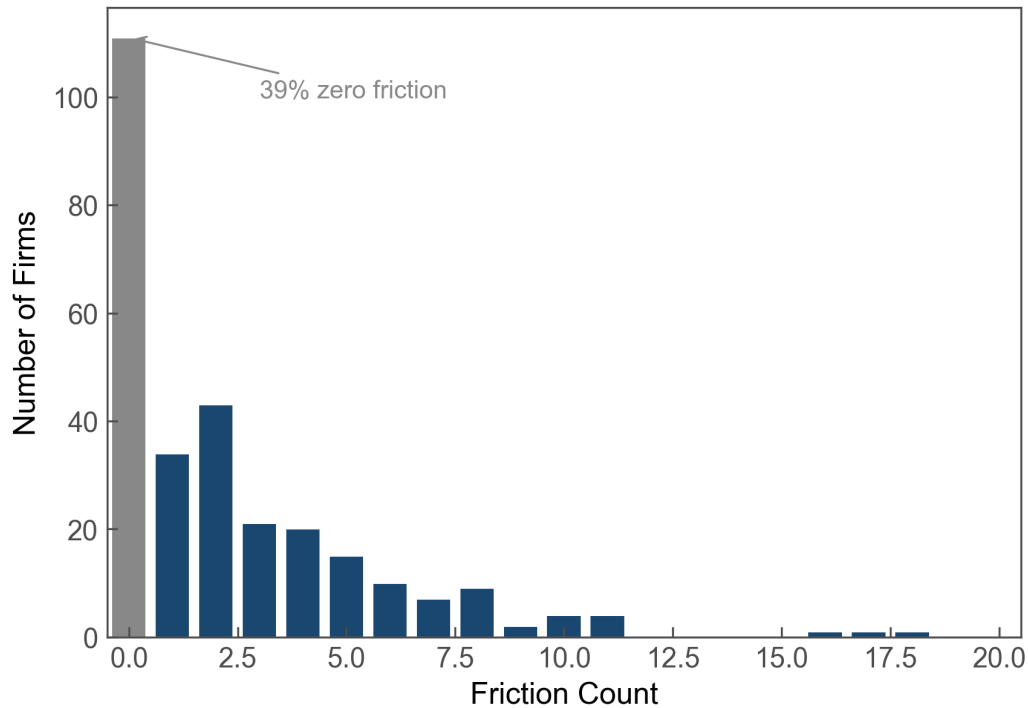


Note: Each point represents a firm at E1. Colour indicates FF3 CAR($-1, +3$), winsorised at the 1st and 99th percentiles. Dashed lines mark sample medians. ρ denotes the Pearson correlation between log market capitalisation and total friction count ($\rho = -0.21$), confirming that the two channels are not collinear.

A.3 Supplementary Figures

A.3.1 Friction Distribution

Figure 8: Distribution of regulatory friction counts across the SGX equity panel. The majority of firms have zero or one friction event; the right tail is driven by firms with repeated queries, halts, and suspensions.



Note: Distribution of total friction event count across the 335 equity-only sample firms. Friction events include SGX queries, trading halts, and suspensions over August 2022 to August 2025.

Figure 8 shows the heavily right-skewed friction distribution that underlies the estimates in Table 8. Because most firms record zero or one friction event, the identifying variation comes chiefly from the contrast between zero-friction and moderate-friction firms.

A.4 Additional Event and Cross-Sectional Results

A.4.1 E0: Review Group Formation

The Review Group formation announcement on August 2, 2024 (E0) produced a strongly negative mean CAR of -2.15% , significant at the 1% level, with only 30.8% of firms posting positive returns. Although the cross-sectional regressions likewise yield significant size and friction coefficients

in line with the patterns documented for E1, the event window coincides with the late-July to early-August 2024 JPY carry trade unwind, an episode that generated large negative returns across Asian equity markets. Once the market model includes a JPY factor (FF4-JPY), the REIT effect and several other cross-sectional patterns disappear, indicating that the E0 results are confounded by the carry trade unwinding. We therefore treat E0 as uninformative for hypothesis testing and rely on E1 as the primary announcement event.

A.4.2 E2 and E3 Detailed Results

The E2 consultation event (May 15, 2025) produced a null mean CAR of -0.11% ($p = 0.721$), and no firm-level variable reaches significance at conventional levels. This null is consistent with the consultation having been fully anticipated once E1 announced that detailed proposals would follow “by mid-2025.” E3 (July 21, 2025) produced a uniformly positive mean CAR of 3.12% , significant at the 1% level, with 70.0% of firms posting positive returns. The continuous size measure reverses sign relative to E1, to a value significant at the 5% level (-0.006), while the friction count is economically zero ($p > 0.55$). The deployment therefore raised returns across the equity panel, with smaller firms gaining the most; the interpretation of this reversal is developed in [Section 5.1](#).

A.4.3 E5: Final Report

The final report event (E5, November 19, 2025) produced a positive mean CAR of 0.62% , significant at the 1% level, with 60.5% of firms posting positive returns. Across firms the above-median size indicator is positive and significant at the 1% level (0.015), and illiquidity carries a positive sign significant at the 5% level (0.007), the opposite direction from E1. Friction count and watch-list status are both insignificant.

A.4.4 Governance Variables

Table [23](#) reports univariate governance regressions across E1, E2, E3, and E4. Board independence share, non-busy board membership, and turnover rate are insignificant at all four events, while CEO-chair separation is insignificant at E1, E2, and E4, reaching marginal significance only at E3 (-0.022). The exception is ‘Fresh independence’ — whether the board appointed at least one new independent director in the preceding fiscal year — which is positive and significant at the 1% level at E3 (0.092) and at the 5% level at E4 (0.019). This pattern suggests that firms with recently refreshed boards benefited more than others from the capital deployment and the deregulation implementation.

Table 23: Governance Variables Across Events

	E1	E2	E3	E4
Independence share	0.0038 (0.0260)	-0.0240 (0.0243)	-0.0113 (0.0375)	-0.0131 (0.0374)
No CEO duality	0.0108 (0.0104)	0.0087 (0.0092)	-0.0222* (0.0126)	0.0015 (0.0096)
Non-busy board	-0.0264 (0.0255)	-0.0016 (0.0252)	-0.0469 (0.0409)	-0.0194 (0.0307)
Fresh independence	-0.0131 (0.0101)	0.0019 (0.0125)	0.0919*** (0.0113)	0.0192** (0.0081)
Turnover rate	-0.0046 (0.0029)	-0.0020 (0.0026)	-0.0019 (0.0053)	-0.0002 (0.0030)

Notes: Univariate OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Governance variables from director data. E4 governance merged from E1 cross-section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4.5 Broker/Institutional Ownership

Singapore’s nominee and custody structure channels large shareholdings through intermediary accounts, making the shareholder register difficult to read at face value. We therefore split the top-20 shareholder register into four categories: insider (directors and substantial shareholders), nominee (bank nominee accounts), broker (retail securities brokers), and institutional (fund managers and sovereign wealth funds). Although nominee accounts may hold shares for any type of investor, broker accounts typically serve retail investors who maintain securities accounts with their brokerage. In this context, we use *Broker %* – the share of top-20 holdings registered through retail securities brokers – as a proxy for ‘retail-dominated ownership’. A high broker share marks a register held mainly by retail investors, with few institutions positioned to trade on news.

The broker-ownership subsample covers 205 of 274 E1 firms (75%). The excluded firms are marginally smaller ($p = 0.082$) and overwhelmingly Catalyst-listed, with the latter composition difference significant at the 1% level (97% vs 81%). CARs and friction counts, by contrast, do not differ significantly between the two subsamples. The ownership-rebalancing sample is narrower, covering the 172 firms with top-20 shareholder data in both FY2024 and FY2025.

Table 24 estimates stepwise OLS regressions of E1 abnormal returns on broker ownership. In column 1, a one percentage point increase in broker ownership is associated, on its own, with a -0.0007 lower CAR, significant at the 1% level. Adding size controls reduces the coefficient to -0.0006 (column 2), significant at the 5% level; adding friction controls then yields -0.0005

(column 3), again significant at the 5% level. The coefficient remains -0.0005 in the fullest specification, which adds financial controls, illiquidity, and Catalist (column 5), and retains marginal significance: broker ownership predicts E1 returns even after size and friction are controlled.

Table 25 reports the same pattern through broker-ownership quartile sorts. At E1, mean CARs fall steadily from -0.36% in Q1 (lowest broker ownership) to -2.37% in Q4 (highest), leaving a $Q4 - Q1$ spread of -2.00 pp that is marginally significant. At E3, the ranking flips: all four quartiles show positive mean CARs, and the $Q4 - Q1$ spread changes sign to 1.70 pp, although it is now statistically indistinguishable from zero. The broker-ownership gradient therefore reverses between E1 and E3, paralleling the size reversal discussed in Section 5.1.

Table 26 asks whether post-reform ownership changes are aligned with the policy's intended channel. If the programme had channelled capital towards small and mid-cap firms, nominee holdings — the primary custodial vehicle for institutional investment — should have risen more for smaller firms. The estimates move in the opposite direction. Controlling for FY2024 nominee levels, larger firms show higher FY2025 nominee ownership (coefficient on $\log(\text{MktCap}) = 1.25$, significant at the 5% level; column 1), while Catalist-listed firms show a 3.5 percentage point decline in broker holdings relative to Mainboard peers, also significant at the 5% level (column 2). These ownership patterns are consistent with the announcement-return evidence that the capital channel favoured larger, more liquid firms, although they do not by themselves isolate the causal effect of the EQDP on ownership structure.

Table 24: Broker Ownership and E1 Abnormal Returns

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Broker ownership and E1 abnormal returns</i>					
Broker %	−0.0007 ^{***} (0.0002)	−0.0006 ^{**} (0.0003)	−0.0005 ^{**} (0.0002)	−0.0006 ^{**} (0.0002)	−0.0005 [*] (0.0003)
Log(Mktcap)		0.0036 (0.0023)	0.0023 (0.0022)	0.0027 (0.0022)	0.0011 (0.0026)
Friction count			−0.0033 [*] (0.0018)	−0.0036 ^{**} (0.0017)	−0.0037 ^{**} (0.0018)
Watchlist				0.0699 [*] (0.0385)	0.0628 (0.0419)
Illiquidity					0.0027 (0.0095)
Catalist					−0.0057 (0.0139)
Financial controls	No	No	No	No	Yes
<i>N</i>	205	204	204	204	192
<i>R</i> ²	0.046	0.059	0.094	0.115	0.147

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is FF3 CAR(−1, +3), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Broker % is the share of top-20 holdings registered through retail securities brokers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: Broker Ownership Quartile Sorts

	E1: February 2025		E3: July 2025	
	Mean	Median	Mean	Median
Q1 (lowest)	-0.36 (0.66)	-0.12	+1.97** (0.93)	+0.76
Q2	-0.45 (0.57)	-0.54	+3.16*** (0.97)	+1.91
Q3	-1.75* (0.97)	-1.52	+4.98*** (1.35)	+3.24
Q4 (highest)	-2.37*** (0.86)	-1.94	+3.68*** (1.36)	+2.85
Q4 - Q1	-2.00* (-1.85)		+1.70 (1.03)	
N		205		201

Notes: Mean and median CARs (%) by broker-ownership quartile for E1 (21 February 2025) and E3 (21 July 2025). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses; significance is assessed by the Student t -test. Stars on mean CARs denote the significance of a one-sample t -test against zero. The $Q4 - Q1$ row reports the difference in means; Welch t -statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 26: Post-E3 Ownership Rebalancing

	(1)	(2)	(3)
Nominee % (FY2024)	0.8369 ^{***} (0.0537)		
Log(Mktcap)	1.2515 ^{**} (0.5484)	−0.2459 (0.2659)	
Catalist	0.2180 (1.6345)	−3.5228 ^{**} (1.5889)	
Medium			−1.0171 (1.6969)
Large			−2.8088 (2.0851)
<i>N</i>	172	172	172
<i>R</i> ²	0.815	0.025	0.015
Large-cap Δ Top-1 %	−2.92 ^{**} [$t = -2.22$, $p = 0.031$]		

Notes: OLS with HC1 robust standard errors in parentheses. Dependent variables are FY2025 nominee ownership (column 1), the FY2024-to-FY2025 change in broker ownership (column 2), and the FY2024-to-FY2025 change in top-1 shareholder ownership (column 3), estimated on the subsample of firms with top-20 shareholder data in both fiscal years. Nominee % (FY2024) is the lagged dependent variable. Medium and Large are size-tercile indicators (omitted category: Small). The final row reports the mean change in top-1 shareholder ownership for large-cap firms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4.6 Realised Dollar Volume Across Reform Sub-Periods

Although the main results measure how the market expected the reform to matter by comparing firms within announcement windows, this appendix asks whether realised trading moved the same way. To answer it, we split the post-E1 period into four sub-periods and compare the size gradient in realised dollar volume with the announcement-window CAR pattern. The exercise is descriptive. Each sub-period spans weeks to months, during which shocks unrelated to the reform occur, and raw dollar volume cannot be netted against a factor-model benchmark. Consequently, the single-date, cross-firm design that identifies the main results cannot be reproduced here.

The sample is a balanced panel of 260 SGX-listed equity issuers (128 above-median and 131 below-median at E1), each contributing at least three trading days in every sub-period. The reference baseline is Pre-E1 (January 2, 2024 to February 14, 2025), which matches the estimation window used in our CAR specifications. Four post-E1 sub-periods are compared against this baseline:

1. E1→E3: February 21 to July 20, 2025 (post-reform announcement, pre-mandate disclosure).
2. E3→E4: July 21 to October 28, 2025 (EQDP active deployment, pre-implementation of E4

reforms).

3. E4→E5: October 29 to November 18, 2025 (post-watch-list abolition, anticipating E5 structural reforms).
4. Post-E5: November 19 to December 31, 2025 (post-implementation of E5).

Separating E3→E4 from E4→E5 isolates the active deployment of EQDP from the deregulation and structural-reform packages introduced at E4 and E5. From E3 (July 21) the named managers are mandated to begin deployment, whereas the E4→E5 window absorbs both the SGX RegCo deregulation implemented at E4 and any market anticipation of the E5 structural package, both of which [Section 3](#) details.

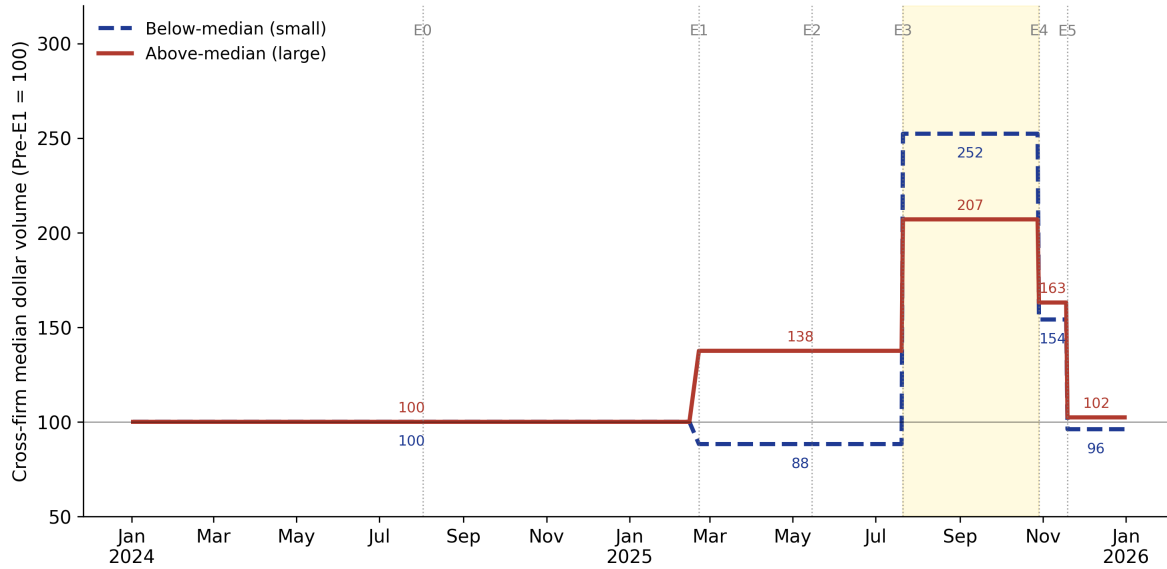
[Table 27](#) reports cross-firm median dollar volume per firm-day for each group and sub-period, together with percent changes from Pre-E1, and [Figure 9](#) indexes each series to its Pre-E1 baseline of 100 and plots the period-level statistic as a step function. Through E1→E3, the below-median series sits below baseline while the above-median series rises. Both groups then jump sharply at E3→E4, where the below-median series reaches an index of 252 against 207 for the above-median group. The gains fade through E4→E5 and Post-E5, and both groups return to within a few percent of baseline by the end of December 2025.

Table 27: Cross-Firm Median Dollar Volume per Firm-Day, by Sub-Period

<i>Period</i>	Below-median (small)		Above-median (large)	
	Median \$/firm-day	Δ% vs Pre-E1	Median \$/firm-day	Δ% vs Pre-E1
Pre-E1 (baseline)	\$40,459	—	\$1,027,699	—
E1→E3	\$35,708	−11.7%	\$1,413,671	+37.6%
E3→E4	\$102,086	+152.3%	\$2,127,791	+107.0%
E4→E5	\$62,353	+54.1%	\$1,675,995	+63.1%
Post-E5	\$38,885	−3.9%	\$1,051,942	+2.4%

Notes: Balanced panel of 260 SGX-listed equity issuers (128 above-median, 131 below-median at the E1 cross-section), each with at least three trading days in every sub-period. Each cell reports the cross-firm median of within-firm mean dollar volume per trading day in the indicated sub-period, with percent change from the Pre-E1 baseline. Pre-E1 spans January 2, 2024 to February 14, 2025 (the CAR estimation window).

Figure 9: Rolling Median Dollar Volume by Group



Notes: Cross-firm median of within-firm period mean dollar volume by group, indexed to Pre-E1 = 100. Each step corresponds to the period-level statistic reported in Table 27. Dotted vertical lines mark the reform events. The shaded region demarcates the E3→E4 sub-period, the only window in the timeline that isolates EQDP active deployment from subsequent reform implementation. Sample: 260 SGX-listed equity issuers.

On the median split, E3→E4 is the only post-E1 window in which volume grew proportionally faster among below-median firms than among above-median firms. In E1→E3, E4→E5, and Post-E5, the ordering reverses. This median ranking is consistent with the deployment phase favouring small-cap trading, but it is descriptive: the panel regression cannot distinguish the two groups within the E3→E4 window, although the above-median advantage there is significantly smaller than in the E4→E5 window that follows.¹²²

The widening gap between larger and smaller firms at E4→E5 and Post-E5 appears only once the later deregulation and structural-reform package enters the relevant window. Three components of that package mechanically favour large firms. First, the SGX–Nasdaq dual-listing bridge, by easing a primary listing on both boards, carries fixed costs that favour larger issuers. Second, the board-lot reduction above S\$10 lowers minimum trade size only for higher-priced stocks, which cluster in larger capitalisations. Third, designated-market-maker rebates raise the expected per-quote revenue of market making in firms where it is already profitable, predominantly mid-caps. These channels operate outside the capital and information channels analysed in our article.

The descriptive evidence adds two refinements to the interpretation of the main results. First, real trading moved with the E3 repricing: dollar volume rose well above baseline in both size groups during the deployment window. Second, the post-E5 widening of the size gap in realised

¹²²A within-firm panel regression of daily $\log(1 + \text{shares traded})$ on $\text{AboveMedian} \times \text{sub-period}$ interactions, with firm and date fixed effects and firm-clustered standard errors, yields coefficients of +0.25 ($p = 0.013$) at E1→E3, +0.13 ($p = 0.45$) at E3→E4, +0.56 ($p = 0.026$) at E4→E5, and +1.18 ($p < 0.001$) Post-E5; a Wald test rejects equality of the E3→E4 and E4→E5 coefficients ($\Delta\hat{\beta} = -0.42$, $p = 0.016$). A continuous-size specification replacing the binary indicator with log market capitalisation yields qualitatively identical results.

trading appears only with the structural package described above, after the deployment window had closed. It therefore does not bear on the capital channel.

Several caveats qualify this evidence. Even within the deployment window, coincident factors, including sector cycles, cannot be ruled out. The two later windows are also thin: E4→E5 spans only about 15 trading days, Post-E5 spans about 30, and the latter overlaps SGX year-end seasonality. The proportional rankings in those windows therefore rest on few observations, while the persistence of the size-gap widening cannot be assessed within this sample. A final limitation is more basic. If the EQDP managers bought and held, their deployment would leave little trace in realised volume. The available institutional-ownership disclosures, which reflect fiscal-year-ends predating E3, cannot resolve whether they did.